

The rules of sex and strain: deciphering the chemical language of inbred and wild mouse conspecific scents

Reviewed Preprint

Revised by authors after peer review.

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Reviewed preprint version 2

April 11, 2024 (this version)

Reviewed preprint version 1

September 1, 2023

Sent for peer review

July 11, 2023

Posted to preprint server

June 26, 2023

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Abstract

Summary

In most mammals, conspecific chemosensory communication relies on semiochemical release within complex bodily secretions and subsequent stimulus detection by the vomeronasal organ (VNO). Urine, a rich source of ethologically relevant chemosignals, conveys detailed information about sex, social hierarchy, health and reproductive state, which becomes accessible to a conspecific via vomeronasal sampling. So far, however, numerous aspects of social chemosignaling along the vomeronasal pathway remain unclear. Moreover, since virtually all research on vomeronasal physiology is based on secretions derived from inbred laboratory mice, it remains uncertain whether such stimuli provide a true representation of potentially more relevant cues found in the wild. Here, we combine a robust low-noise VNO activity assay with comparative molecular profiling of sex- and strain-specific mouse urine samples from two inbred laboratory strains as well as from wild mice. With comprehensive molecular portraits of these secretions, VNO activity analysis now enables us to (i) assess whether and, if so, how much sex- / strain-selective “raw” chemical information in urine is accessible via vomeronasal sampling; (ii) identify which chemicals exhibit sufficient discriminatory power to signal an animal’s sex, strain, or both; (iii) determine the extent to which wild mouse secretions are unique; and (iv) analyze whether vomeronasal response profiles differ between strains. We report both sex- and, in particular, strain-selective VNO representations of chemical information. Within the urinary ‘secretome’, both volatile compounds and proteins exhibit sufficient discriminative power to provide sex- and strain-specific molecular fingerprints. While total protein amount is substantially enriched in male urine, females secrete a larger variety at overall comparatively low concentrations. Surprisingly, the molecular spectrum of wild mouse urine does not dramatically exceed that of inbred strains. Finally, vomeronasal response profiles differ

between C57BL/6 and BALB/c animals, with particularly disparate representations of female semiochemicals.

eLife assessment

This carefully executed study provides a comparison of the chemical composition of mouse urine across strain and sex with the responses of vomeronasal sensory neurons, which are responsible for detecting chemical social cues. While the authors did not examine all molecular classes found in mouse urine or directly test whether the urinary volatile chemicals that vary with sex and strain are effective vomeronasal neuron ligands, **solid** data are provided that will be of significant interest to those studying chemical communication in rodents. This work should provide a **valuable** foundation for future research that will determine which molecules drive sex- and strain-specific vomeronasal responses.

Introduction

In rodents and most other mammals, the accessory olfactory system controls conspecific chemical communication during social interactions ^{1,4}. Ethologically meaningful chemostimuli that trigger stereotypic social and sexual behaviors are predominantly detected by the vomeronasal organ (VNO), a tubular sensory structure at the anterior base of the nasal septum. In a crescent-shaped medial neuroepithelium, the VNO harbors approximately 100,000 to 200,000 vomeronasal sensory neurons (VSNs ⁵), each extending a single unbranched apical dendrite that terminates in a paddle-shaped swelling ⁴. At the dendritic tips, microvilli are immersed in mucus that fills a central luminal canal, which extends via the vomeronasal duct into the nasal cavity. During social investigatory behavior, which in mice primarily involves periods of intense licking and sniffing of both facial and anogenital regions ⁶, semiochemicals are sucked into the VNO lumen. Upon binding to cognate vomeronasal receptors (VRs), the chemical signal is transduced into electrical VSN activity and, ultimately, neuronal discharge.

Behaviourally relevant natural chemostimuli are, typically, complex blends of compounds ⁷ in various bodily secretions ⁸. By far, the most widely studied secretion in animal chemosensory research is urine (see ⁴ and references therein), which is a rich source of semiochemicals that serves a well-established function in social communication. While we still lack a comprehensive molecular description of this broadband vomeronasal stimulus, previous work has identified several putative semiochemicals in mouse urine and other bodily secretions, which activate VSNs and cover many structural groups and feature dimensions ^{7,9,18–21,10–17}. Prominent molecularly identified VSN stimuli include various sulfated steroids ^{9,21–26}, which could reflect the dynamic endocrine state of an individual.

So far, virtually all research on vomeronasal physiology is based on urinary stimuli derived from inbred laboratory mice (but see ²⁷). This facilitates standardization across studies but, while VSN sex- and strain-selectivity has previously been explored ^{21,28}, it remains unclear whether secretions collected from inbred mice accurately represent the potentially more ethologically relevant stimuli found in the wild. In fact, while wild *Mus musculus* populations exhibit several-fold higher levels of genetic variation than those in human populations, commonly studied inbred strains of laboratory mice derive from a limited set of founders. Therefore, such strains contain only a small subset of the genetic variation that is present in nature ²⁹. Classical inbred strains of house mice are genetic mosaics of the three main wild subspecies, *M. m. domesticus*, *musculus*, and *castaneus*, which started to diverge ~350–500 thousand years ago. However, the genomes of

prominent inbred strains, such as C57BL/6 or BALB/c, are predominantly derived from *M. m. domesticus*³⁰. The mitochondrial genomes of both laboratory strains are identical, implying a common descent along the maternal line. Given their overlapping geographical distribution, wild house mice subspecies have also undergone secondary contact and hybridization³¹, which has diversified their genetic landscape.

Laboratory inbreeding, which has been ongoing for more than 100 years for C57BL/6 and BALB/c strains²⁹, may have affected the chemical composition of bodily secretions and, consequently, their information content. Indeed, both within and between strains, laboratory mice lack variation in major urinary protein (MUP) patterns typically found among wild mice³². This inbred homogeneity could have important implications for research investigating, for example, social recognition or mate choice: if there are marked qualitative and/or quantitative differences in chemical composition of secretions and, accordingly, in the neuronal representations of wild as compared to inbred derived stimuli, one might question the conclusions based on a large body of work using inbred secretions²⁷.

On the recipients' end, the level of chemosensory information available from con-/heterospecific secretions is determined by individual VR expression profiles. Across strains, VR repertoires vary as a function of both genetics and experience^{33–36}. Therefore, it is unclear whether receptor arrays in laboratory strains, despite hundreds of generations of inbreeding and domestication (“microevolution”) in a laboratory environment, have retained selectivity for more ethologically relevant wild-derived stimuli. Moreover, it remains uncertain whether VR tuning profiles enable VSNs to capture key ethological features from molecular concentration differences between sex- and strain-specific secretions^{15,37}. Together, four key biological questions arise from these lines of reasoning: (i) Do VSN response profiles reflect the global molecular content of urine (i.e., are these neurons sensitive to many/most compounds) or, by contrast, is the VNO a highly selective molecular detector (responding to just a few select molecules)? (ii) Are VSN response profiles strain-specific? (iii) Which semiochemicals provide information about sex and/or strain? (iv) Is there something unique about wild mouse secretions and/or the VSN response profiles they trigger?

Here, to address these unresolved issues, we combine a robust VSN activity assay with comparative molecular profiling of sex- and strain-specific mouse urine from two inbred laboratory strains and wild mice. Our study provides molecular portraits of these secretions. We report that large fractions of generic urine compounds are shared among both male and female mice of all genetic backgrounds. We further show that the urinary ‘secretome’ in wild mice does not differ dramatically from that found in laboratory strains. Surprisingly, while male urine contains much higher protein concentrations, females secrete a larger variety of proteins (including MUPs). For proteins common to all strains, concentrations are relatively low in C57BL/6, moderate in BALB/c, and high in wild animals. However, despite this concentration bias, there is no overrepresentation of wild-selective proteins. Notably, both the volatile organic compound (VOC) and protein profile of urine, each provides sufficient information to decode a given sex/strain combination (with protein content exhibiting stronger discriminative power). Moreover, we identify a rich lipocalin repertoire in urine, which alone could allow chemosensory discrimination of sex/strain combinations.

A key strength of this study is the use of the exact same stimuli as previously employed to investigate sensory representations in the accessory olfactory bulb (AOB)²⁷, the first central processing stage along the accessory olfactory system pathway. Our previous work demonstrated that AOB representations of ethologically relevant urine stimuli are similar for male mice from two different inbred strains (C57BL/6 and BALB/c), despite potential differences in VR repertoires. In addition, we found that wild mouse stimuli elicit responses that, although not identical, are nevertheless qualitatively similar to those from commonly used inbred strains²⁷. VSN activity analysis now enables us to ask whether these features also manifest in the VNO and, thus, to assess

(i) whether the information inherent in a sex- and strain-specific urinary secretome is accessible via vomeronasal sampling; and (ii) if, on the population level, VSN sensory representations differ between strains. Comparing responses from male C57BL/6 and BALB/c mice (as done previously on AOB level [27](#)), our data demonstrate that recipient strain identity is reflected by their VSN activity patterns. Moreover, when exposed to stimuli from different strains, large VSN fractions (often >50 %) respond to only one stimulus, suggesting a substantial degree of selective sampling. Together, our study reveals selective and strain-dependent representations of urine chemical content, a surprising scarceness of selective responses to wild stimuli, as well as remarkably rich and sex-/strain-specific molecular profiles that likely preserve most of the biologically relevant information.

Results

In mice, the VNO is the predominant sensory structure mediating conspecific chemical communication. In this study, by comparing samples derived from two inbred strains (BALB/c and C57BL/6) as well as from wild mice, we pursue four main questions: (i) Which chemical components in mouse urine – a major source of semiochemicals – distinguish sex and/or strain? (ii) To what degree has inbreeding affected the chemical composition of urine (i.e., how unique are wild mouse secretions)? (iii) How much of the sex- and strain-specific chemical information in urine is accessible to a conspecific via vomeronasal sampling (i.e., how selective are VSN response profiles)? (iv) Upon exposure to the same stimuli, do VSN response patterns differ between strains?

A “low noise” assay to capture VSN population activity

To record VSN signal fingerprints in response to naturalistic stimuli, it is essential to establish a robust population activity assay that reliably captures the raw information content inherent in bodily secretions. To this end, we analyzed single-cell Ca^{2+} transients among large VSN populations in acute coronal VNO sections (**Figure 1a,b**). Our experimental design followed a standard pairwise comparison paradigm (**Figure 1c**). Pairs of pooled stimuli differed either in the donors' sex (male *versus* female) or in their genetic background (BALB/c, C57BL/6, or wild). We repeated brief (10 s) alternating stimulus presentations twice at 180 s inter-stimulus intervals that ensured recovery from VSN adaptation [38](#). Experiments concluded with a brief exposure to elevated extracellular K^+ (S_3) to depolarize neurons and test for integrity of each neuron's spike generation machinery. Dependent on their individual urine response profiles, VSNs were categorized as either specialists (selective response to one stimulus) or generalists (responsive to both stimuli). Overall, we recorded >43,000 K^+ -sensitive neurons, of which a total of 16,715 VSNs (38.4%) responded to urine stimulation. Of these urine-sensitive neurons, 61.4% displayed generalist profiles, whereas 38.6% were categorized as specialists (**Figure 1c,d**). As a measure of intrinsic signal variability, we calculated reliability indices that quantify the similarity (or lack thereof) of two successive responses to the same stimulus (with small values reflecting high reliability; see Materials and Methods). We analyzed both Ca^{2+} signal amplitudes (**Figure 1d**) and integrals (**Figure 1 – figure supplement 1a**) as measures of response magnitude. For both generalist and specialist VSNs, response reliability indices are normally distributed around zero. While the distribution of reliability indices derived from response integral measurements is somewhat broad (**Figure 1 – figure supplement 1a**), indices based on signal amplitudes proved relatively homogeneous (**Figure 1d**). Accordingly, we use average response amplitudes as indicators of signal strength throughout this study.

In a first set of control experiments, we asked how VSNs from male C57BL/6 mice respond when challenged with urine samples from two groups of animals of the same sex and inbred strain (**Figure 1 – figure supplement 1b,c**). Since the chemical composition of both stimuli should be similar, we expected that the vast majority of urine-sensitive VSNs will display generalist response

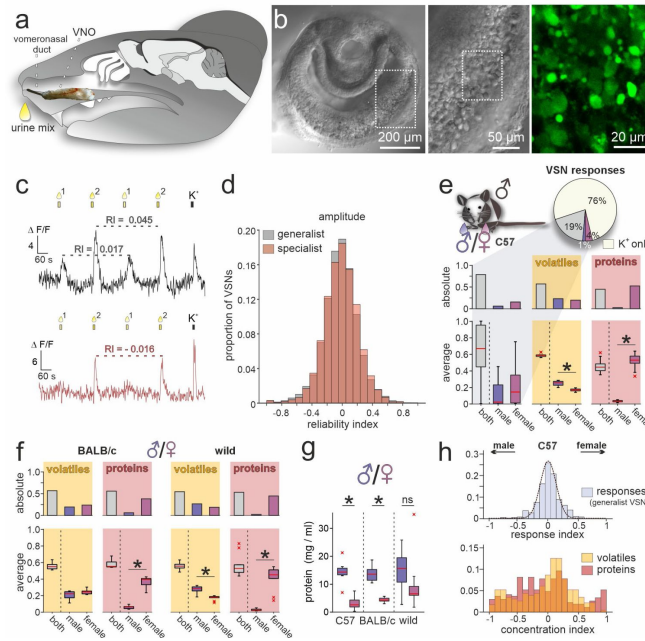


Figure 1

A population activity assay captures sex-dependent stimuli representations.

(a) Anatomical location of the rodent VNO. Schematic depicting a sagittal section through a mouse head with an overlay of a VNO image. The vomeronasal duct that opens into the anterior nasal cavity is also highlighted. (b) Overview (left) and zoomed-in (middle) differential interference contrast micrograph showing an acute coronal VNO section from an adult C57BL/6 mouse. Confocal fluorescence image (right; dashed rectangle in middle) depicting a depolarization (elevated K^+) dependent cytosolic Ca^{2+} increase in VSN somata after bulk loading with Cal-520 AM. (c) Original traces showing changes in cytosolic Ca^{2+} concentration over time in two representative VSN somata. VNO slices were briefly challenged with two mixtures of diluted mouse urine (1:100; 10s; top yellow bars/droplets). Repeated stimulation in alternating sequence at 180 s inter-stimulus intervals was followed by membrane depolarization upon exposure to elevated extracellular K^+ (50 mM; 10 s). According to individual response type, VSNs were categorized as ‘generalists’ (top) or ‘specialists’ (bottom). Signal amplitudes in response to the same stimulus allowed calculation of a reliability index (RI) as a measure of signal robustness. (d) Amplitude reliability index histograms of all generalist VSNs (gray; $n = 10,258$) and all specialist VSNs (red; $n = 6,457$) recorded in this study. Note that for both response types, indices are normally distributed with a narrow central peak around zero. (e) Quantification of results obtained from recordings in VSNs from male C57BL/6 mice challenged with male versus female C57BL/6 urine stimuli. Pie chart (top) illustrates the proportions of generalist (19%) and specialist neurons (1% and 4%, respectively) among all K^+ -sensitive VSNs ($n = 1,999$). Bar graph (middle, left) breaks down the summed total of urine-sensitive neurons by categories and compares their distribution with the proportions of VOCs (yellow background; $n = 405$) and proteins (red background; $n = 601$) found either in both male and female urine (grey bars) or exclusively in samples from one sex (male, dark blue; female, purple). Box-and-whisker plots (bottom, left) illustrating generalists-to-specialists ratios over individual experiments ($n = 19$). Boxes represent the first-to-third quartiles. Whiskers represent the 10th and 90th percentiles, respectively. Outliers (1.5 IQR; red x) are plotted individually. The central red band represents the population median ($P_{0.5}$). Results are shown in relation to box-and-whisker plots that outline chemical content data obtained from paired comparisons ($n = 10$ individuals per group). (f) Molecular composition of male versus female urine from BALB/c and wild mice. Bar graphs (top) display proportions of VOCs (yellow background; BALB/c, $n = 514$; wild, $n = 462$) and proteins (red background; BALB/c, $n = 407$; wild, $n = 526$) found either in both male and female urine (grey bars) or exclusively in samples from one sex (male, dark blue; female, purple). Box-and-whisker plots (bottom) quantify category ratios for individual paired experiments ($n = 10$ individuals per group). (g) Quantification (Bradford assay) of protein/peptide content in urine samples from C57BL/6, BALB/c, and wild mice, respectively ($n = 10$ each). Note the substantially increased protein content in male samples. (h) Response index histogram (top) obtained from generalist C57BL/6 VSNs that responded to both male and female same-strain urine (data corresponding to (e)). Fitted Gaussian curve (dashed line) centers close to zero (peak = -0.03) and shows a relatively narrow width ($\sigma = 0.13$). By contrast, concentration index histograms (bottom), calculated for VOCs (yellow) and proteins (red) found in both male and female urine samples, are heterogeneous and not normally distributed. Asterisks (*) indicate statistical significance, $p < 0.05$; Wilcoxon signed-rank test (for VSN functional data), Mann-Whitney U test (for molecular profiling), and unpaired t-test (for protein content comparison shown in (g)).

profiles. That was indeed the case. Less than 2% of VSNs showed stimulus selectivity (**Figure 1 – figure supplement 1b,c**). Moreover, response response indices (reflecting a bias toward either of the paired stimuli) are normally (and narrowly) distributed around zero (**Figure 1 – figure supplement 1b,c**). Both observations confirm a low level of biological and experimental noise in this setting. Thus, our assay is well-suited to detect and compare VSN sensory responses upon pairwise stimulation.

Sex-specific stimuli elicit distinct VSN sensory representations

We next investigated how sex differences are reflected in VSN response profiles. When challenged with male *versus* female urine from C57BL/6 animals, the fraction of specialist neurons more than doubled to 5% of all K^+ -sensitive neurons. Notably, female urine recruited more specialist neurons than male urine (**Figure 1e**). We then asked whether this pattern is correlated with the chemical compositions of male and female urine. In-depth molecular analysis of urine content via GCxGC-MS as well as nLC-MS/MS identified a total of 1006 molecules (detected in ≥ 3 out of 10 male or female samples, respectively), of which approximately 40% are low molecular weight VOCs, while 60% are proteins. Roughly half of the molecules in either group are found in both male and female urine (**Figure 1e**). Unexpectedly, while we hardly identify any male-specific proteins in C57BL/6 urine, we find a large fraction of female-specific proteins. We asked whether this phenomenon is (i) a distinct feature of C57BL/6 mice, (ii) common among inbred laboratory strains, or (iii) also observed in wild animals. Therefore, we included urine samples from both BALB/c and wild mice in extended molecular profiling. In both groups, we observed similarly increased levels of female-specific proteins (**Figure 1f**). The total amount of protein, however, is substantially enriched in male urine (**Figure 1g**). Our data thus suggest that, while females secrete a larger variety of proteins, overall concentrations are comparatively low.

Finally, we asked whether common compounds (i.e., molecules identified in both male and female urine) show concentration disparities between sexes and, if so, how such differences are reflected in VSN response profiles. When VSNs were challenged with male *versus* female C57BL/6 urine, 19% of all K^+ -sensitive neurons responded to both stimuli (**Figure 1e**). Here, again, response preference indices are narrowly distributed and center around zero (**Figure 1h**). By contrast, analogously calculated concentration indices (see Materials and Methods) that can reflect potential disparities are distributed more broadly and non-normally (**Figure 1h**). This apparent incongruence between nonpreferential generalist sensitivity and relatively large chemical concentration differences indicates that VSN responses (at least in C57BL/6 males) do not simply reflect the information theoretically available via concentration differences between male and female same-strain stimuli. One explanation for this is that common ligands are not well-represented by the observed general concentration disparities. Alternatively, it may be that even low concentrations fully activate a given VSN, and thus concentration differences are not reflected by response strengths. Notably, a broad and non-normal distribution of concentration indices among molecular components of male *versus* female urine is also found in samples from both BALB/c and wild mice (**Figure 1 – figure supplement 1d**).

Chemical characterization of urine content identifies distinctive molecular signatures of sex and strain

Which differences in chemical composition (i.e., which molecules) characterize sex-/strain-specific secretions and may thus provide information about sex and strain to a recipient? Molecular profiling of urine content provides a unique opportunity to address this question and identify enriched compounds (i.e., present in ≥ 6 of 10 individual samples) with the potential capacity to signal an animal's sex, strain, or both. Adopting this more conservative criterion, we detected 208 enriched VOCs and 264 enriched proteins. Almost half of all VOCs (47.6%) and about one-third of all proteins (31.4%) are found in all six sex/strain combinations tested and are, thus, considered generic mouse urine components (**Figure 2a,b**). Moreover, a total of 33 compounds (10 VOCs, 23 proteins) were identified across all strains, but in a sex-specific fashion. Similarly, 22 molecules (10

VOCs, 12 proteins) revealed strain selectivity, independent of sex. Notably, a large fraction of compounds, i.e., 34.1% of all VOCs (71 / 208) and 22.3% of proteins (59 / 264), were detected exclusively in one of the six sex/strain combinations. Only 18 of the 208 VOCs (8.7%) could not be categorized as either generic or specific (for either sex, or strain, or a unique sex/strain combination). Among the 82 proteins that showed sex specificity, either across strains (23 proteins) or as part of a unique sex/strain combination (59 proteins), the vast majority (89.0%) was found in female samples. For VOCs, however, the opposite picture emerged. Here, 52 of 81 sex-specific compounds (64.2%) were selectively detected in male samples. Together, chemical profiling revealed (i) that large fractions of urine content are shared among both laboratory and wild mice; (ii) that roughly one-third of urinary VOCs and one-fourth of proteins are exclusively found in a given sex/strain combination; and (iii) that male and female mice might have adopted different chemical secretion strategies to signal their sex.

If information coding along the accessory olfactory pathway would strictly follow a ‘labeled-line’ logic³⁹, absence or presence of a given molecule could be adequate to signal sex or strain. Several recent lines of evidence, however, suggest a combinatorial coding strategy that also involves some level of circuit plasticity^{4,34,40}. Therefore, we next asked if the total VOC or protein content of a given urine sample preserves sufficient predictive / discriminative information to classify samples according to sex, strain, or a specific sex/strain combination. We used sparse Partial Least-Squares Discriminant Analysis (sPLS-DA)⁴¹, a chemometrics machine learning technique, to reduce data dimensionality and optimize sample separation⁴². When plotted on two-dimensional coordinates that represent the most discriminative variables (**Figure 2c-h**), both VOCs (**Table 1**) and proteins provide sufficient information to cluster stimuli according to sex (**Figure 2c,f**), strain (**Figure 2d,g**), or a combination of both variables (**Figure 2e,h**). We then calculated Variable Importance in Projection (VIP) scores as measures of a particular variable’s informative power, which is correlated to the variance explained by the model^{41,42}. Generally, we find that protein content exhibits stronger discriminative power (21-24% & 11-12% of explained variance) than VOC content (7-10% & 5-8% of explained variance).

We next aimed to identify the most relevant variables (i.e., molecules) for sex or strain classification. Training Random Forest classifiers⁴³, we obtained feature significance scores, the “Gini importance”⁴³, that provide relative relevance rankings of the individual variables. Supplementary Figure 2 lists the 20 most informative VOCs (**Table 2**) and proteins that discriminate sex and strain, respectively (**Figure 2 – figure supplement 1a,b**), along with the abundance of the corresponding top 5 molecules (VOCs or proteins) for each of the six sex/strain combinations (**Figure 2 – figure supplement 1c-f**). Notably, major urinary protein 20 (Mup20 / darcin)⁴⁴, fatty acid-binding protein 5 (Fabp5)⁴⁵, and N-acetylgalactosamine-6-sulfatase (Galns)⁴⁶ exhibit substantial power to discriminate sex across strains (**Figure 2 – figure supplement 1b,f**). While Mup20 and Galns are considerably more abundant in male urine, Fabp5 appears to be specific for female samples. Strain discrimination, on the other hand, is optimal with protein-tyrosine kinase 2-beta (Ptk2b)⁴⁷, RNase T2 (Rnaset2)⁴⁸, prosaposin (Psap)⁴⁹, lymphocyte antigen 6A-2/6E-1 (Ly6a)⁵⁰, and superoxide dismutase 1 (Sod1)⁵¹ (**Figure 2 – figure supplement 1b,d**). Specifically, Ptk2b is detected exclusively in C57BL/6 mice. Ly6a is absent in BALB/c animals, whereas Rnaset2 is largely missing in both C57BL/6 and wild mice. With proteins exhibiting stronger discriminative power than VOCs (**Figure 2c-h**), future studies will have to focus on these proteins to identify potential functions as vomeronasal chemosignals.

Overall, we have detected similar amounts of proteins / VOCs across strains. However, individual variability in protein and VOC content was significantly higher in wild mice than in both laboratory strains (**Figure 2 – figure supplement 2a**). We confirmed secretion of previously reported putative semiochemicals^{10,11,15,40,52}, including both VOCs (**Figure 2 – figure supplement 2b**) and proteins (**Figure 2 – figure supplement 2c**) in both wild and laboratory mice. Notably, however, all Mups (including Mup20) and most such VOCs were found in samples

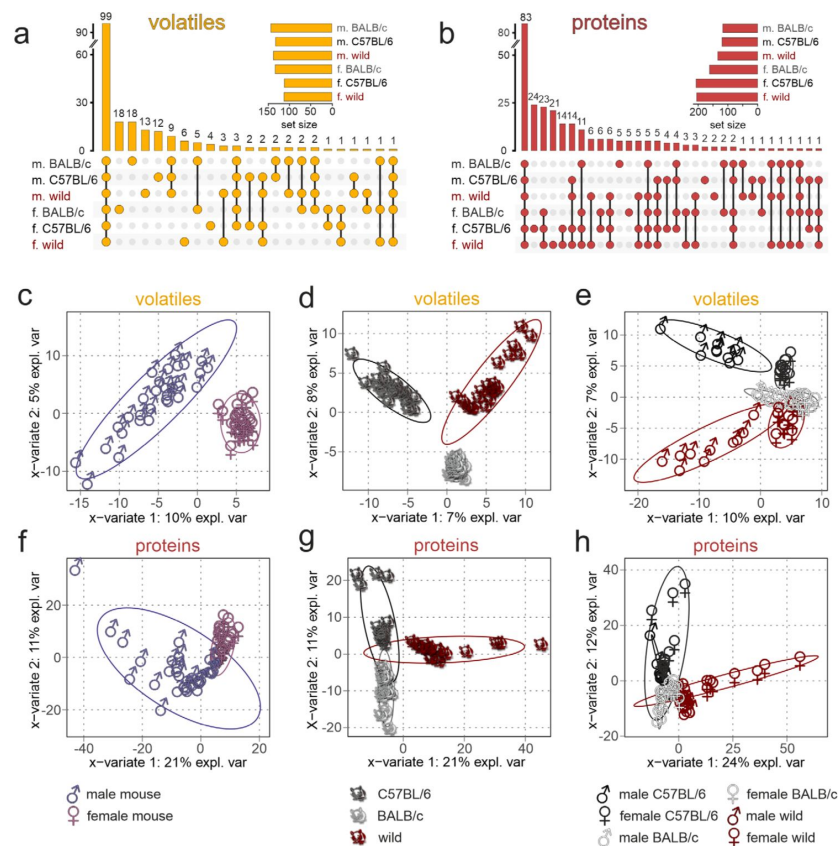


Figure 2

Chemical profiling of urine content identifies unique sex- and strain-specific molecular fingerprints.

(a&b) Matrix layout for all intersections of VOCs (a) and proteins (b) among the six sex/strain combinations, sorted by size. Colored circles in the matrix indicate combinations that are part of the intersection. Bars above the matrix columns represent the number of compounds in each intersection. Empty intersections have been removed to save space. Horizontal bar charts (bottom, left) depict the number of VOCs (a) and proteins (b) detected in each urine set. Proteomics data are available via ProteomeXchange with identifier PXD042324. (c-h) Sparse Partial Least-Squares Discriminant Analysis (sPLS-DA) score plots depicting the first two sPLS-DA components, which explain 7-10% (1st component) and 5-8% (2nd component) of VOC data variance (c-e) as well as 21-24% (1st component) and 11-12% (2nd component) of protein data variance (f-h), respectively. Ellipses represent 95% confidence intervals. Plots demonstrate sample clustering according to the urine donors' sex (c&f), genetic background (d&g), or sex/strain combination (e&h). Each data point represents a sample from an individual animal (n = 60; 10 samples per sex/strain combination), with sample type colored according to symbol legend (bottom).

int ID	MS-DB-ID	bias / sex	comp	Chem-ID	alt. name	ChEBI / CID	ori
A140	Methane, nitro-	male	1	CH ₃ NO ₂	Nitrocarbol	77701	?
A336	Thiazole, 2-ethyl-4,5-dihydro-	male	1	C ₅ H ₉ NS	2-Ethylthiazoline	86896	rodents
A220	Butanenitrile, 2-methyl-	male	1	C ₅ H ₉ N	2-Cyanobutane	51937	insects
A305	2-sec-Butylthiazole	male	1	C ₇ H ₁₁ NS	2-(1-Methylpropyl)thiazole	519539	mouse
A209	2-Butenal, 2-methyl-	male	1	C ₅ H ₈ O	Tiglic aldehyde	88419	mammals
A278	Furan, 2,3-dihydro-4-(1-methylethyl)-	male	1	C ₇ H ₁₂ O	Furan	35559	mammals
A102	2-Butanone	female	1	C ₄ H ₈ O	Butan-2-one	28398	mammals
A497	Benzyl alcohol	female	1	C ₇ H ₈ O	Phenylmethanol	17987	mammals
A28	Acetaldehyde	female	1	C ₂ H ₄ O	Ethanal	15343	mammals
A248	Butyric acid, 3-tridecyl ester	female	1	C ₁₇ H ₃₄ O ₂	Tridecyl butyrate	169401	rodents
A307	1-Octen-3-ol	female	1	C ₈ H ₁₆ O	1-Vinylhexanol	34118	ubiquitous
A765	1,2-Benzenedicarboxylic acid, butyl 2-methylpropyl ester	female	1	C ₁₆ H ₂₂ O ₄	Butyl isobutyl phthalate	519539	mammals
int ID	MS-DB-ID	bias / strain	comp	Chem-ID	alt. name	ChEBI / CID	ori
A51	Trichloromethane	C57BL/6	2	CHCl ₃	Chloroform	6212	ubiquitous
A14	Cyclopropane, ethyl-	C57BL/6	2	C ₅ H ₁₀	Ethylcyclopropane	70933	ubiquitous
A74	Ethyl Acetate	C57BL/6	2	C ₄ H ₈ O ₂	Acetoxyethane	8857	yeast
A10	Ethylenimine	C57BL/6	2	C ₂ H ₅ N	Aziridine	9033	ubiquitous
A123	n-Propyl acetate	wild	2	CH ₃ COOCH ₂ CH ₂ CH ₃	Propyl ethanoate	40116	insects
A355	Pyrazine, 2-ethenyl-6-methyl-	wild	2	C ₇ H ₈ N ₂	Pyrazine, 2-methyl-6-vinyl-	518838	ubiquitous
A659	Dodecanoic acid, isoctyl ester	wild	2	C ₂₀ H ₄₀ O ₂	Isooctyl laurate	2216918	plants
A165	2-Pentanone	wild	2	C ₅ H ₁₀ O	Ethyl acetone	7895	ubiquitous
A372	Pyridine, 2,3,4,5-tetrahydro-	wild	1	C ₅ H ₉ N	Piperidine	47858	ubiquitous
A486	Benzenamine, 3-methyl-	wild	1	C ₆ H ₄ CH ₃ NH ₂	3-Toluidine	7934	?
A156	Pentanal	BALB/c	2	C ₅ H ₁₀ O	Valeraldehyde	8063	insects
A181	Isopropylsulfonyl chloride	BALB/c	2	C ₃ H ₇ ClO ₂ S	2-Propanesulfonyl chloride	82408	bacteria
A456	Cyclohexane, isothiocyanato-	BALB/c	2	C ₇ H ₁₁ NS	Isothiocyanocyclohexane	14289	?
A45	Heptane, 2,4-dimethyl-	BALB/c	2	C ₉ H ₂₀	2,4-dimethylheptane	16656	insects
A265	Benzene, 1,2,3-trimethyl-	BALB/c	1	C ₉ H ₁₂	Hemellitrol	1797279	insects
A251	Benzene, 1,2,4-trimethyl-	BALB/c	1	C ₉ H ₁₂	Pseudocumol	7247	insects
A977	Propene	BALB/c	1	CH ₂ CHCH ₃	Methylethylene	6378	?

Table 1

List of VOCs that – according to sPLS-DA – display the most discriminative power to classify samples by sex (top; related to **Figure 2c**) or strain (bottom; related to **Figure 2d**). Note that for all VOCs that facilitate sex discrimination, x-variate 1 is most informative. VOCs that are potentially important for strain differentiation, however, are best separated by either x-variate 1 or 2. Entries list internal mass spectrometry identifiers (int ID), identifiers extracted from MS analysis database (MS-DB-ID), the sex or strain that drives separation (bias), which two-dimensional component / x-variate represents the most discriminative variable (comp), PubChem chemical formula (Chem-ID), PubChem common or alternative name (alt. name), Chemical Entities of Biological Interest (ChEBI) or PubChem Compound Identification (CID), and putative origin (ori).

int ID	MS-DB-ID	bias / sex	Chem-ID	alt. name	ChEBI / CID	ori
A140	Methane, nitro-	male	CH ₃ NO ₂	Nitrocarbol	77701	?
A220	Butanenitrile, 2-methyl-	male	C ₅ H ₉ N	2-Cyanobutane	51937	insects
A235	3-Penten-2-one	male	C ₅ H ₈ O	(E)-pent-3-en-2-one	637920	insects
A336	Thiazole, 2-ethyl-4,5-dihydro-	male	C ₅ H ₉ NS	2-Ethylthiazoline	86896	rat
A13	Methanethiol	male	CH ₃ SH	MTMT	878	mammals
A20	Methylamine, N,N-dimethyl-	male	C ₃ H ₉ N	Trimethylamine	1146	mouse
A765	1,2-Benzenedicarboxylic acid, butyl	female	C ₁₆ H ₂₂ O ₄	Butyl isobutyl phthalate	519539	mammals
A240	2-Penten-1-ol, (Z)-	male	C ₅ H ₁₀ O	trans-2-Pentenol	5364920	insects
A209	2-Butenal, 2-methyl-	male	C ₅ H ₈ O	Tiglic aldehyde	88419	mammals
A305	2-sec-Butylthiazole	male	C ₇ H ₁₁ NS	2-(1-Methylpropyl)thiazole	519539	mouse
A337	2-sec-Butylthiazole (2nd variant)	male	C ₇ H ₁₁ NS	2-(1-Methylpropyl)thiazole	519539	mouse
A102	2-Butanone	female	C ₄ H ₈ O	Butan-2-one	28398	mammals
A152	2,4-Dimethyl-1-hexene	male	C ₈ H ₁₆	2,4-Dimethylhex-1-ene	519301	?
A618	6-Methyl-2-pyridinecarbaldehyde	male	C ₇ H ₇ NO	6-Methylpicolinaldehyde	70737	?
A84	4-Hexen-3-one, 5-methyl-	male	C ₇ H ₁₂ O	5-Methylhex-4-en-3-one	256081	insects
int ID	MS-DB-ID	bias / strain	Chem-ID	alt. name	ChEBI / CID	ori
A51	Trichloromethane	C57BL/6	CHCl ₃	Chloroform	6212	ubiquitous
A265	Benzene, 1,2,3-trimethyl-	BALB/c	C ₉ H ₁₂	Hemellitrol	1797279	insects
A63	Acetone	C57BL/6	C ₃ H ₆ O	Dimethyl ketone	180	ubiquitous
A14	Cyclopropane, ethyl-	C57BL/6	C ₅ H ₁₀	Ethylcyclopropane	70933	ubiquitous
A172	(S)-(+)-2-Pentanol	wild	C ₅ H ₁₂ O	2-Pentanol	22386	mouse
A222	2-Hexanone	wild	C ₆ H ₁₂ O	Hexanone	11583	mouse
A165	2-Pentanone	wild	C ₅ H ₁₀ O	Ethyl acetone	7895	ubiquitous
A251	Benzene, 1,2,4-trimethyl-	BALB/c	C ₉ H ₁₂	Pseudocumol	7247	insects
A123	n-Propyl acetate	wild	C ₅ H ₁₀ O ₂	Propyl acetate	7997	insects
A355	Pyrazine, 2-ethenyl-6-methyl-	wild	C ₇ H ₈ N ₂	Pyrazine, 2-methyl-6-vinyl-	518838	mammals
A74	Ethyl Acetate	C57BL/6	C ₄ H ₈ O ₂	Acetoxyethane	8857	yeast
A156	Pentanal	BALB/c	C ₅ H ₁₀ O	Valeraldehyde	8063	insects
A146	Cyclohexene,3-(1-methylpropyl)-	wild	C ₁₀ H ₁₈	3-Sec-butyl-1-cyclohexene	139917	insects
A194	o-Xylene	wild	C ₆ H ₄ (CH ₃) ₂	Ortho-Xylene	7237	?
A83	Benzene	wild	C ₆ H ₆	Benzole	241	insects

Table 2

List of VOCs that – according to Random Forest classification and resulting Gini importance scores – display the most discriminative power to classify samples by sex (top; related to **Figure 2 - figure supplement 1a**) or strain (bottom; related to **Figure 2 - figure supplement 1b**). As for several compounds listed in **Table 1**, many VOCs that display high relative rankings of individual variable relevance are common metabolites. Generally, it is reassuring that several VOCs are listed in both **Table 1** and **Table 2**, emphasizing that two different supervised machine learning algorithms (i.e., sPLS-DA (**Table 1**) and Random Forest (**Table 2**)) yield largely congruent results. Here, entries (in blue if identified by both sPLS-DA and Random Forest) list internal mass spectrometry identifiers (int ID), identifiers extracted from MS analysis database (MS-DB-ID), the sex or strain that drives separation (bias), PubChem chemical formula (Chem-ID), PubChem common or alternative name (alt. name), Chemical Entities of Biological Interest (ChEBI) or PubChem Compound Identification (CID), and putative origin (ori).

from either sex, albeit at male-biased concentrations for Mup3, Mup17, and Mup20 (**Figure 2 – figure supplement 2c** [↗](#)). The only compounds showing male-specific secretion are 2-sec-butyl-4,5-dihydrothiazole (SBT) and farnesenes (**Figure 2 – figure supplement 2b** [↗](#)), which have previously been implicated as facilitators of female mouse puberty acceleration [11](#) [↗](#).

Notably, we find a rich repertoire of 27 lipocalins in mouse urine. When based exclusively on lipocalin content, hierarchical clustering groups individual samples into a set of clusters that, with very few exceptions (5 out of 60), correspond to the six sex/strain combinations (**Figure 2 – figure supplement 3** [↗](#)). This finding, therefore, demonstrates the power of urinary lipocalins for potential chemosensory discrimination of sex and/or strain.

VSNs are more selective for strain than for sex

Next, we challenged neurons with stimulus pairs from two same-sex / different-strain combinations and asked whether VSN response profiles reflect the molecular fingerprint of corresponding urine samples (e.g., male C57BL/6 *versus* male BALB/c (**Figure 3a** [↗](#))). We analyzed a total of 17,416 K⁺-sensitive VSNs (**Figure 3a-f** [↗](#)). Again, we distinguished between specialist VSNs that responded exclusively to one stimulus, and generalists, which responded to both stimuli. Along the same lines, we categorized either strain-specific or broadly detected VOCs and proteins. Several conclusions emerge from these classifications: (i) with one exception (i.e., upon stimulation with male C57BL/6 *versus* wild stimuli (**Figure 3b** [↗](#))), roughly half of all urine-sensitive VSNs are generalists – a result consistent with our finding that generally ~50% of urine molecules are shared among compared strains. Accordingly, the fraction of strain-selective (specialist) responses is considerably larger than observed for sex-specific responses (**Figure 1e** [↗](#)); (ii) in female urine, BALB/c-specific proteins are substantially underrepresented, a fact not reflected by VSN response profiles (**Figure 3d,f** [↗](#)); (iii) surprisingly, the amount of strain-specific molecules in wild mouse urine does not vastly exceed that in inbred strains; and (iv) accordingly, selective VSN responses to wild stimuli are by no means more common (**Figure 3b,c,e,f** [↗](#)).

Pronounced strain-dependent concentration imbalances between common urinary compounds are not reflected by generalist VSNs

When comparing generalist VSN responses to male *versus* female C57BL/6 urine (**Figure 1e,h** [↗](#)) we noted that the narrow normally distributed stimulus response index histograms did not match the broader and heterogeneous distributions of concentration disparities between sexes (**Figure 1h** [↗](#); S1b-d). For sex-dependent cues, as mentioned above, this could indicate that semiochemical concentration differences carry only limited information. Next, we therefore asked whether strain-specific concentration differences between urine samples exist and, if so, whether such differences can convey information about strain. In total, we recorded strain-independent generalist responses from 3,366 VSNs in male C57BL/6 mice (**Figure 4** [↗](#)). In all but one experimental condition (i.e., when comparing male stimuli from BALB/c *versus* wild mice) response preference indices were normally distributed and well fit by relatively narrow Gaussian curves that centered around zero. By contrast, chemical analysis revealed that many compounds identified in urine from both strains differ substantially in concentration. For proteins, in particular, strong concentration disparities exist in all strain combinations analyzed. In both male and female samples, concentrations of common proteins are relatively low in C57BL/6, moderate in BALB/c, and high in wild animals. As shown in **Figures 2b** [↗](#) and **3** [↗](#), this concentration bias towards wild proteins does not translate into any dramatic overrepresentation of proteins selectively found in wild male and/or female urine. In fact, the massively skewed distributions of protein concentration indices are not reflected by generalist VSN profiles. The latter better match VOC concentration distributions, which generally display broad, yet Gaussian shapes. We conclude that VSN population response strength might not be so strongly affected by strain-dependent concentration differences among common urinary proteins. In that case, it would appear somewhat unlikely that individual VSN activity provides fine-tuned information about distinct

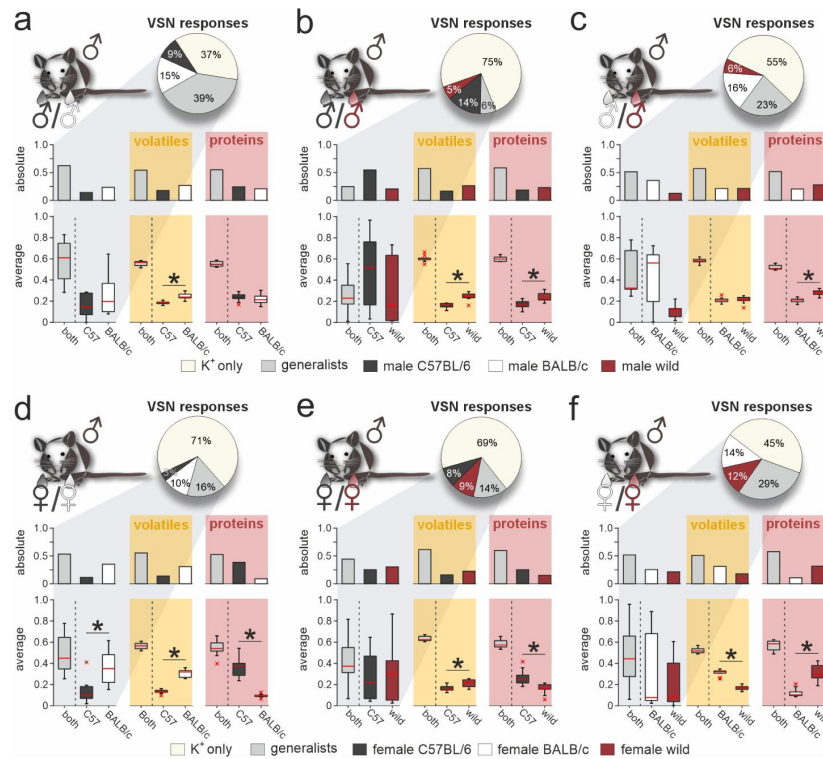


Figure 3

Selective VSN response profiles upon exposure to strain-specific signatures.

Quantitative comparison between VSN responses to paired stimuli and their respective chemical signatures. Neurons of male C57BL/6 mice were challenged with male (**a-c**) or female (**d-f**) urine, respectively. Response profiles are compared upon stimulation with C57BL/6 *versus* BALB/c urine (**a&d**), C57BL/6 *versus* wild stimuli (**b&e**), and BALB/c *versus* wild urine (**c&f**), respectively. Pie charts (top) illustrate the proportions of generalist (light gray) and specialist neurons (dark gray, white, and purple, respectively) among all K⁺-sensitive VSNs (**a**, n = 1855; **b**, n = 4116; **c**, n = 2462; **d**, n = 2376; **e**, n = 3230; **f**, n = 3377). Bar graphs (left) break down the urine-sensitive neurons by categories and compare their distribution with the proportions of VOCs (middle; yellow background; **a**, n = 450; **b**, n = 448; **c**, n = 479; **d**, n = 480; **e**, n = 405; **f**, n = 492) and proteins (right; red background; **a**, n = 317; **b**, n = 330; **c**, n = 334; **d**, n = 657; **e**, n = 715; **f**, n = 584) found either in both urine types (gray bars) or exclusively in samples from one group (color code as in pie charts). Box-and-whisker plots (bottom) illustrate generalist-to-specialist VSN category ratios over individual experiments (**a**, n = 6; **b**, n = 10; **c**, n = 7; **d**, n = 9; **e**, n = 10; **f**, n = 12). Boxes represent the first-to-third quartiles. Whiskers represent the 10th and 90th percentiles, respectively. Outliers (1.5 IQR; red x) are plotted individually. The central red band represents the population median (P_{0.5}). Results are shown in relation to box-and-whisker plots that outline chemical content data obtained from paired comparisons (n = 10 individuals per group). Asterisks (*) indicate statistical significance, *p* < 0.05, Wilcoxon signed-rank test.

semiochemical concentrations. Alternatively, as some (or even many) of the identified proteins could not serve as vomeronasal ligands at all, generalist VSNs might sample information from only a subset of compounds which, in fact, are secreted at roughly similar concentrations.

Vomeronasal representation of female semiochemicals differs between two inbred strains

So far, our approach was restricted to VSN signals recorded from male C57BL/6 mice. An important question, of course, is whether the response profiles we observed are themselves recipient strain-dependent. Thus, we next aimed to assess the extent to which our findings generalize to VSN populations from another laboratory animal strain. We therefore repeated all six pairwise same-sex / different-strain stimulation experiments, using acute VNO slices from male BALB/c mice (**Figures 5** & **S5**). Categorization as generalist or specialist VSNs revealed that proportions differed significantly between VSNs from BALB/c *versus* C57BL/6 mice, albeit at varying degrees. Similar to C57BL/6 neurons, selectivity of BALB/c neurons to wild-derived stimuli was rather rare. In fact, (i) compared to an average specialist rate of $11.2\% \pm 6.6\%$ (mean $\pm$ SD) calculated over all 13 binary stimulus pairs ($n = 26$ specialist types), we recorded hardly any specialist responses upon stimulation with urine from wild females (2% and 3%, respectively), and (ii) we found comparatively few generalist signals when wild-derived female urine was among the paired stimuli (**Figure 5e,f**). This striking insensitivity is not observed in VSNs from C57BL/6 mice, suggesting substantial differences in VR expression between the two inbred laboratory strains. Notably, we can rule out that the observed differences result from differential sampling along the VNO's anterior-to-posterior axis since, for each experiment, we routinely sample and average across slices cut along the organ's entire anterior-to-posterior length (see Materials and Methods).

Next, we examined whether generalist VSN response profiles differ between male BALB/c and C57BL/6 animals (**Figure 5 – figure supplement 1**). In total, we recorded generalist responses from 1,741 BALB/c neurons. When plotting response preference indices, we noticed less homogeneous distributions than previously observed in C57BL/6 mice. Five of the six histograms showed multiple peaks and thus could not be fitted by a single Gaussian, whereas the only histogram adhering to a normal distribution was comparatively broad (**Figure 5 – figure supplement 1a**). As observed in C57BL/6 neurons, the skewed distributions of protein concentration indices were not reflected by BALB/c generalist VSN profiles. Comparison of generalist VSN response histograms between BALB/c and C57BL/6 mice (**Figure 5 – figure supplement 1c,d**) revealed strong and consistent differences upon stimulation with female stimuli (**Figure 5 – figure supplement 1d**). While single Gaussians around zero characterize C57BL/6 generalist distributions upon stimulation with female stimuli (**Figure 5 – figure supplement 1d**), several prominent peaks emerged when fitting histograms derived from BALB/c VSNs. Notably, for some generalist BALB/c neurons, wild-derived female stimuli are less potent than their inbred strain counterparts. This finding either indicates reduced concentrations of the corresponding molecules in wild urine (rendering most proteins unlikely candidates (**Figure 5 – figure supplement 1a,b**)), or suggests some yet to be determined form of cooperativity upon receptor-ligand interaction. Together, for both generalist and specialist VSNs, vomeronasal representation of female semiochemicals differs considerably between the two inbred mouse strains.

In the present study, we have established a robust VSN activity assay that allows pairwise comparison of neural selectivity and response strength upon stimulation with chemically defined natural stimuli. In-depth chemical analysis of sex- and strain-specific individual urine samples revealed that (i) large fractions of urine content are shared among mice of all sex / strain combinations; that (ii) the amount of molecules selectively found in wild mouse urine does not dramatically exceed the urinary secretome of inbred strains; that (iii) across strains, female-

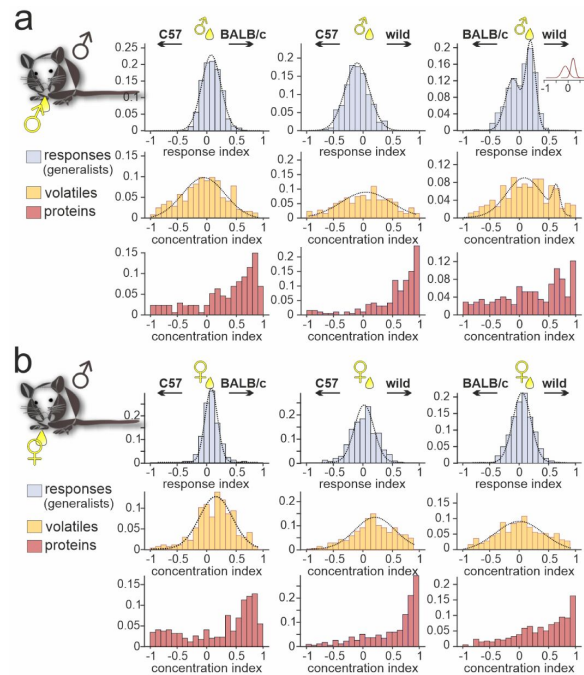


Figure 4

Strain-dependent concentration imbalances exert relatively mild effects on VSN population response homogeneity.

Comparison of male C57BL/6 generalist VSN response preferences, upon stimulation with paired urine stimuli from different strains, with strain-dependent concentration imbalances among VOCs and proteins, respectively. **(a&b)** Response index histograms (top rows) depict distributions of generalist data outlined in **Figure 3** (gray bars). With one exception (**a** (right), male BALB/c *versus* male wild), histograms are well fitted by single Gaussian curves (dashed lines) that each center relatively close to zero (**a** (left), peak = 0.08, $\sigma = 0.18$; **a** (middle), peak = -0.11, $\sigma = 0.21$; **a** (right), 1st peak = -0.12, 1st $\sigma = 0.14$; 2nd peak = 0.21, 2nd $\sigma = 0.09$; **b** (left), peak = 0.07, $\sigma = 0.12$; **b** (middle), peak = 0.03, $\sigma = 0.23$; **b** (right), peak = 0.05, $\sigma = 0.18$). Concentration index histograms (middle & bottom rows), calculated for VOCs (yellow) and proteins (red) found in both tested urine samples, are more heterogeneous. Notably, while most VOC concentration index histograms are also fitted by single, albeit broader Gaussian curves (**a** (left), peak = -0.07, $\sigma = 0.37$; **a** (middle), peak = 0.02, $\sigma = 0.45$; **a** (right), 1st peak = 0.07, 1st $\sigma = 0.32$; 2nd peak = 0.64, 2nd $\sigma = 0.06$; **b** (left), peak = 0.15, $\sigma = 0.29$; **b** (middle), peak = 0.21, $\sigma = 0.35$; **b** (right), peak = -0.01, $\sigma = 0.42$), protein concentration imbalances are not normally distributed.

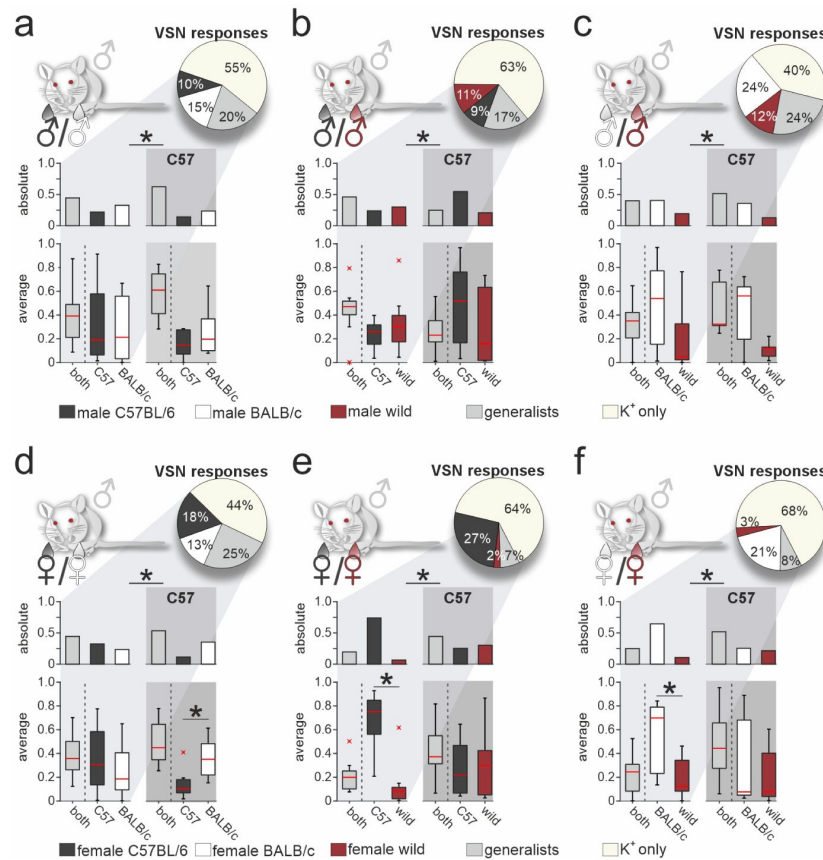


Figure 5

Vomeronasal representation of female semiochemicals differs between inbred strains.

Comparison of VSN response profiles between male BALB/c and C57BL/6 mice. Pie charts (top) illustrate the proportions of generalist (light gray) and specialist BALB/c neurons (dark gray, white, and purple, respectively) among all K⁺-sensitive VSNs (**a**, $n = 1244$; **b**, $n = 2063$; **c**, $n = 1903$; **d**, $n = 1770$; **e**, $n = 1934$; **f**, $n = 1818$). VSNs were challenged with male (**a-c**) or female (**d-f**) urine, respectively. Response profiles are compared upon exposure to C57BL/6 *versus* BALB/c urine (**a&d**), C57BL/6 *versus* wild stimuli (**b&e**), and BALB/c *versus* wild urine (**c&f**), respectively. Bar graphs break down the urine-sensitive neurons by categories and compare distributions among BALB/c neurons (left) to responses recorded from C57BL/6 VSNs (right; gray background). Box-and-whisker plots (bottom) illustrate generalist-to-specialist VSN category ratios over individual experiments (**a**, $n = 8$; **b**, $n = 10$; **c**, $n = 9$; **d**, $n = 10$; **e**, $n = 9$; **f**, $n = 10$). Boxes represent the first-to-third quartiles. Whiskers represent the 10th and 90th percentiles, respectively. Outliers (1.5 IQR; red x) are plotted individually. The central red band represents the population median ($P_{0.5}$). Asterisks (*) indicate statistical significance, $p < 0.05$, Wilcoxon signed-rank test.

specific proteins vastly outnumber the male-specific variety, while (iv) overall protein concentration is substantially enriched in male urine; that (v) concentrations of common proteins are relatively low in C57BL/6, moderate in BALB/c, and high in wild animals; that (vi) both secreted VOC and protein profiles provide sufficient information to distinguish sex, strain, or both; and that (vii) the rich urinary lipocalin repertoire alone might allow chemosensory discrimination of sex and/or strain.

When asking how much of this chemical information is accessible to inbred male mice via vomeronasal sampling we observe that (i) VSN population response profiles do not reflect the global molecular content of urine, suggesting that the VNO functions as a rather selective molecular detector; that (ii) selective VSN responses to wild stimuli are by no means more common (in fact, selectivity to wild-derived stimuli is rather rare); that (iii) VSN generalist signal strength is unlikely to encode semiochemical concentrations across the entire range of compounds; that (iv) male BALB/c neurons display striking insensitivity when challenged with urine from wild females; and, thus, that (v) vomeronasal representation of female semiochemicals differs considerably between inbred strains.

Discussion

Urine is the primary source of social chemosignals among mice (and, in fact, many other mammals) and contains both ‘fixed’ (i.e., genomic) information about strain, sex, individual identity, genetic histocompatibility and background, as well as ‘variable’ (i.e., metabolic) information on current social, reproductive and health status⁵³. The ability to glean ethologically meaningful information from chemosensory sampling of urine (or any conspecific bodily secretion) depends on (i) the specific (semio)chemical composition of a urine sample, and (ii) the sensory apparatus used for sampling. For most mammals, the VNO is the key chemosensory structure involved in detecting conspecific chemical cues⁴. The virtually universal use of inbred laboratory mice in research aimed at understanding VNO physiology – as both the donors of stimuli and the experimental subjects employed in these studies – could have resulted in misconceptions and biased notions about vomeronasal signaling and, thus, conspecific chemical communication. If that were the case, one might question the relevance and ethological validity of conclusions drawn from a large body of work using inbred secretions²⁷. Here, we addressed this issue from both a chemical ecology and a physiological perspective. In-depth comparative molecular profiling of urine from two classical laboratory strains as well as wild mice reveals several shared features, but also qualitative and quantitative differences in composition. Furthermore, we observe substantial differences in vomeronasal representations of stimuli between C57BL/6 and BALB/c sensory neuron populations.

For analytical purposes, we separate the urinary ‘volatilome’ and proteome. In chemosensory research, this distinction has often been conceptualized as general (i.e., airborne) odors, which activate the main olfactory system, *versus* vomeronasal stimuli⁵⁴. This notion, however, is misleading since organic compounds with low molecular weight and high vapor pressure (i.e., VOCs) in bodily secretions do not instantly evaporate, of course. Rather, they are readily accessible for vomeronasal sampling upon direct contact during investigatory behavior. Notably, our chemical profiling approach omits (sulfated) steroids other non-volatile small organic molecules, which have previously been identified in mouse urine as VSN stimuli⁹. Caution should thus be exerted to not attempt to fully explain VSN response specificity based on VOC and protein content alone.

For both urinary VOCs and proteins, large fractions are shared among both male and female mice of either genetic background (**Figure 2a & b**). Such compounds could be considered generic (mouse) urine components and might not even serve any chemosensory signaling functions. Notably, both the urinary volatilome and proteome on their own, each entail sufficient

information to discern an individual's sex and strain, with protein content exhibiting stronger discriminative power. The protein that is most informative for discriminating between sexes is, perhaps not surprisingly, Mup20 (darcin) ⁴⁴. This well-described “maleness signal” had previously been reported to elicit innate attraction and generate a conditioned place preference in females ^{52,55}, whereas, in males, Mup20 promotes aggression ⁴⁰. Chemosensory roles of the second and third best protein determinants of sex discrimination are basically unexplored. Fabp5 ⁴⁵ and Galns ⁴⁶ are substantially enriched in female and male urine, respectively. Fatty acid-binding proteins, including Fabp5, are evolutionary conserved intracellular lipid chaperones that coordinate cellular lipid trafficking and signaling and are thus linked to metabolic and inflammatory pathways ⁴⁵. N-acetylgalactosamine-6-sulfatase (Galns) on the other hand is a lysosomal hydrolase. Five urinary proteins – Ptk2b ⁴⁷, Rnaset2 ⁴⁸, Psap ⁴⁹, Ly6a ⁵⁰, and Sod1 ⁵¹ – display pronounced strain-dependent differences in concentration. None of these has previously been attributed a chemosensory function. Challenging the mouse VNO with purified recombinant protein(s) will help elucidate whether such functions exist.

Proteomic profiling revealed three additional, rather unexpected findings: First, while male urine contains much higher protein concentrations, females of a given strain secrete a larger variety of proteins (including MUPs). In fact, we do not find a single protein that is exclusively detected in males across strains. By contrast, 23 urinary proteins, while present in all three strains, are found only in females. In line with these observations, only one additional Mup (i.e., Mup21) made the list of the 20 most informative proteins that discriminate sex. In general, our data provide little evidence for a sparse molecular code of (fe)maleness. Rather, the concept of ‘signature mixtures’ ^{7,56,57}, which emphasizes a combinatorial ratio code instead of mere presence/absence phenomena, gains traction. Second, we identify a surprisingly rich lipocalin repertoire in urine, which alone could allow chemosensory discrimination of sex/strain combinations. Within a total of 27 lipocalins, individual patterns allow hierarchical clustering into sex/strain-specific groups (**Figure 2 – figure supplement 3**). These data, thus, support the notion that the lipocalin ‘code’, if relevant, is combinatorial. Third, regarding its molecular spectrum, the urinary ‘secretome’ in wild mice does not differ dramatically from the repertoire found in laboratory strains. To paraphrase this generally reassuring conclusion: while inbreeding could have dramatically modified the nature of chemical secretions and, consequently, their perception by other mice, inbred chemostimuli are largely representative of (potentially) ethologically more relevant wild chemosignals. Notably, however, while mean VOC and protein concentrations show similar distributions across sex/strain combinations, individual variability is strongly increased in wild mouse urine (**Figure 2 – figure supplement 2a**). Accordingly, this finding confirms previous reports of increased individual variation in wild mice ^{32,58}. Another inherent factor that could account for differences in urine secretions among each of the groups, and particularly for comparisons between inbred and wild stimuli, is the microbiome ^{27,59}. Yet, we stress that all urine donors were housed in the same facility and fed with the same diet ²⁷.

Some limitations of our study need to be acknowledged. Regarding stimuli, the six secretion sets we used do not cover the entire coding capacity of the accessory olfactory system ²⁶. Moreover, for VSN response profiling, stimulus samples were pooled across ten individuals in each of the six sex/strain categories. While pooling stimuli reduces individual variability across samples (e.g., regarding fluctuating physiological states), relevant stimulus aspects could be masked. Thus, given the increased chemical variability we observed among individual wild urine samples, pooling might obscure distinctive molecular features of wild mouse secretions. The same holds true for estrus cycle-dependent female stimuli. Because we did not monitor the estrus stage of female urine donors, mixes likely contain samples across the entire cycle. This is relevant as VSN responses may be affected by a donor's cycle stage ^{27,60,61}. Another limitation stems from the use of male inbred mice as experimental subjects. The rationale behind this experimental strategy is to allow for comparisons with our previous study on AOB response profiles ²⁷, which used the exact same settings (see below). Nonetheless, future efforts will have to reveal (i) whether our main findings apply to female recipients; and (ii) if our observations generalize to other

inbred, outbred, or wild mouse strains. As we recently declared [27](#), the latter considerations underscore the importance of recording from wild recipients. While this endeavor presents significant practical challenges, it remains an important goal for future studies.

A surprising, but overall reassuring observation is that responses to wild and inbred stimuli are qualitatively similar. Strikingly, and somewhat counterintuitively, selective VSN responses to wild stimuli are rather rare. This does not result from a generally reduced compound content in wild urine as molecular profiling revealed comparably rich chemical portfolios in wild and inbred samples alike. Rather, we speculate that inbreeding over hundreds of generations in laboratory settings [29](#) has resulted in “microevolutionary” pressure to maintain sensitivity to signals from same- or similar-strain individuals. Indeed, compared to the C57BL/6 reference genome, genetic variability within VR repertoires is massively increased among wild-derived mice (particularly of the *M. m. musculus* subspecies) [36](#). Among the approximately 200 orthologous receptor genes compared, BALB/c genes display 184 non-synonymous and just one private (i.e., unique to a given strain) single-nucleotide polymorphism (SNP). By contrast, VR genes of the wild-derived *M. m. musculus* PWK/PhJ strain show 789 non-synonymous SNPs and 508 private SNPs [36](#).

Initially, we asked whether VSN response profiles reflect the global molecular content of urine or, by contrast, if the VNO serves as a rather selective semiochemical detector. Our findings support a high level of selectivity. When challenged with same-sex / different-strain stimuli, large VSN fractions selectively respond to just one stimulus. Intriguingly, this fraction of strain-selective specialists is considerably larger than observed for sex-specific responses. In line with the notion of highly selective vomeronasal sampling is our observation that the concentration differences between compounds shared among strains, which are often substantial, are not reflected by similarly pronounced differences in response strength among generalist VSNs. There are several, not necessarily mutually exclusive explanations for this finding: First, concentration could simply not be a read-out parameter for VSNs, which would support previous ideas of concentration-invariant VSN activity [15](#). Second, the concentrations in freshly released urine could just exceed the dynamic tuning range of VSNs since, particularly for VOCs, natural signals (e.g., in scent marks) must be accessible to a recipient for a prolonged amount of time (sometimes days). A similar rationale could explain the increased protein concentrations in male urine, since male mice use scent marking to establish and maintain their territories and urinary lipocalins serve as long-lasting reservoirs of VOCs [62](#). Third, generalist VSNs might sample information only from a select subset of urinary compounds, which, given their role as biologically relevant chemosignals, might be released at tightly controlled (and thus similar) concentrations. In fact, in the most extreme scenario, several compounds that do display substantial strain- and/or sex-specific differences in concentration might not act as chemosignals at all. Forth, to some extent, different response profiles could be attributed to non-volatile small organic molecules such as steroids [9](#), which were beyond the focus of our chemical analysis.

While, compared to wild-derived mice, the genetic differences in VR repertoires between C57BL/6 and BALB/c animals appear rather modest [36](#) (see above), vomeronasal representation of female semiochemicals differs considerably between both inbred strains. We conclude that, even in closely related inbred mice, strain-to-strain VR variation must be prominent, an idea supported by various reports of differences in genetic VR makeup across strains [63–65](#). With monoallelic VR expression and 184 described non-synonymous SNPs between C57BL/6 and BALB/c receptor genes, it is likely that even individuals of the same strain express functionally different arrays of VSNs. Adding another layer of complexity, state- and experience-dependent changes in VSN sensitivity have recently been described [34, 66](#).

By adopting the same experimental design (i.e., using the same sets of stimuli in male C57BL/6 versus BALB/c mice) in both this study on VSN response profiles and our previous analysis of sensory representations in the AOB [27](#), we provide a unique comparative perspective on signal transformation along the initial processing nodes of the accessory olfactory pathway. We observe

several differences in representations of ethologically relevant urine stimuli between the VNO and AOB. Notably, while stimulus representations across the two inbred recipient strains were very similar in AOB recordings²⁷, we here observe clear differences in VSN activity, particular upon stimulation with female stimuli. Moreover, a substantial fraction of AOB neurons were selective to wild rather than inbred stimuli, whereas relatively few VSNs showed such selectivity. Consistent with the elaborate wiring patterns within the AOB, these observations imply the presence of non-trivial transformations between VSN and AOB representations. Understanding the exact nature, purpose, and neuronal substrates of these transformations remains an important topic for future studies.

Acknowledgements

This work was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – 368482240/GRK2416 (MS); 378028035 (YBS & MS) – by the Volkswagen Foundation (I/83533, MS), and the German-Israeli Foundation for Scientific Research and Development (1-1193-153.13/2012, YBS & MS). We thank Corinna H. Engelhardt, Stefanie Kurth, and Jessica von Bongartz (RWTH-Aachen University) for excellent technical assistance. We are grateful to Pavel Talacko and Petr Zacek for Proteomic Core Facility support (BIOCEV, Charles University).

Author contributions

Conceptualization, M.Na., P.S., Y.B.-S., and M.S.; Formal Analysis, M.Na. and P.S.; Methodology, M.Na., D.F., R.S., P.S., and M.S.; Investigation, M.Na., M.Ni., R.B., and R.S.; Writing – Original Draft, M.Na. and M.S.; Writing – Review & Editing, M.Na., R.B., D.F., A.L., P.S., Y.B.-S., and M.S.; Funding Acquisition, Y.B.-S. and M.S.; Project Administration, P.S., Y.B.-S., and M.S.; Resources, M.Na., P.S., and M.S.; Supervision, P.S., Y.B.-S., and M.S.; Visualization, M.Na., P.S., A.L., and M.S.

Declaration of Interests

The authors declare no competing interests.

Methods

Animals

All animal procedures were approved by local authorities at RWTH Aachen University, were performed in accordance with local Animal Care and Use Committees' regulations, and in compliance with European Union legislation (Directive 2010/63/EU) and recommendations by the Federation of European Laboratory Animal Science Associations. C57BL/6 and BALB/c mice (Charles River Laboratories, Sulzfeld, Germany) were housed in groups of both sexes [room temperature (RT); 12:12 h light-dark cycle; food and water available *ad libitum*]. All Ca²⁺ imaging experiments used slices from young male adults.

Urine collection from two strains of inbred mice (C57BL/6NCrI and BALB/cAnNCr) as well as first-generation offspring of wild mice was performed at Charles University (Prague, Czech Republic) according to the institute's ethical committee guidelines. Inbred mice were purchased from pathogen-free facilities of the Institute of Molecular Genetics (Czech Academy of Sciences in Prague). Wild *M. m. musculus* mice were caught in house shelters and agricultural buildings near

Prague (Czechia), transferred to the local animal facility at Charles University, and bred for one generation. 20 male (M) and 27 female (F) wild mice were caught at six different sites in the broader Prague area (i.e., Bohnice (50.13415N, 14.41421E; 2M+4F), Dolni Brezany (49.96321N, 14.4585E; 3M+4F), Hodkovice (49.97227N, 14.48039E; 5M+6F), Písnice (49.98988N, 14.46625E; 3M+6F), Lhota (49.95369N, 14.43087E; 1M+2F), and Zalepy (49.9532N, 14.40829E; 6M+5F). 18 of the 27 wild females were caught pregnant. The remaining 9 females were mated with males caught at the same site and produced offspring within a month. All mice that served as urine donors were fed on the same diet. Food and water for all strains were provided *ad libitum* under stable conditions (13:11 h light-dark cycle; 23°C).

Chemicals and solutions

The following solutions (**S**₁–**S**₆) were used:

(**S**₁) 4-(2-Hydroxyethyl)piperazine-1-ethanesulfonic acid (HEPES) buffered extracellular solution containing (in mM) 145 NaCl, 5 KCl, 1 CaCl₂, 1 MgCl₂, 10 HEPES, pH 7.3 (adjusted with NaOH), 300 mOsm (adjusted with glucose).

(**S**₂) Oxygenated (95% O₂, 5% CO₂) extracellular solution containing (in mM) 125 NaCl, 25 NaHCO₃, 5 KCl, 1 MgSO₄, 1 CaCl₂, 5 N,N-bis(2-hydroxyethyl)-2-aminoethanesulfonic acid (BES); pH = 7.3; 300 mOsm (adjusted with glucose).

(**S**₃) Elevated extracellular K⁺ solution containing (in mM) 100 NaCl, 50 KCl, 1 CaCl₂, 1 MgSO₄, 10 HEPES; pH = 7.3 (adjusted with NaOH); 300 mOsm (adjusted with glucose).

If not stated otherwise, chemicals were purchased from Sigma (Schnelldorf, Germany). Solutions and stimuli were applied from air pressure-driven reservoirs via an eight-in-one multibarrel “perfusion pencil” (Science Products, Hofheim, Germany). Changes in focal superfusion⁶⁷ were software-controlled and synchronized with data acquisition by transistor–transistor logic input to 12 V DC solenoid valves using a TIB 14S digital output trigger interface (HEKA Elektronik, Lambrecht/Pfalz, Germany).

Stimuli

For all three donor types (C57BL/6NCrI, BALB/cAnNCr, and wild mice), we collected fresh urine by gentle bladder massage from 10 adult male and female individuals, respectively (resulting in a total of 60 individual samples), starting at an age of 10 weeks. When selecting wild individuals (first-generation offspring) for urine collection, we ensured that all six capture sites (see above) were represented and that age-matched animals displayed similar weight (~17g). To minimize concentration differences that might result from sample-to-sample volume (i.e., dilution) variability, we collected and pooled four to six samples from each individual over several days until each of the 60 animals had provided a total urine volume of >500 µl. Next, we measured general protein content for each sample (Bradford assay). Aliquots of 10 µl were subjected to two-dimensional gas chromatography–mass spectrometry (GCxGC-MS) and nanoliquid chromatography–tandem mass spectrometry (nLC-MS/MS; see below).

The remaining samples were divided into ready-to-use aliquots and stored at –86°C. Prior to experiments, aliquots were thawed and diluted 1:100 in **S**₁⁶⁸. For each of the six strain-/sex-specific stimulus combinations, we created pools from all 10 individuals to minimize individual-to-individual variability. For both inbred and wild female mice, estrus stage was not determined. However, urine collection over several days and pooling across 10 individuals in each stimulus set is designed to reduce variability. Notably, the urine samples employed in this study are the same stimuli used previously to compare sensory representations of inbred and wild stimuli in the AOB of male C57BL/6 and BALB/c mice²⁷.

Two-dimensional gas chromatography-mass spectrometry (GCxGC-MS)

Urine VOCs were sampled with Headspace Solid Phase Micro Extraction (HS SPME) on fiber (DVB/CAR/PDMS grey; Supelco, USA) after 5 min incubation at 55 °C. Next, VOCs were analyzed using two-dimensional gas chromatography with mass detection (Pegasus 4D, LECO Europe B.V., Geleen, The Netherlands) with a combination of mid-polar and non-polar separation columns [primary column: SLB-IL60 (30 m x 0.25 mm, SigmaAldrich, USA); secondary column Rxi-5sil MS (1.4 m x 0.25 mm, Restek, Australia)]. Parameters were set as follows: inlet temperature 270 °C, splitless injection mode, constant He flow 1 ml min⁻¹, modulation time 4 s (hot pulse 0.6 s), modulation temperature offset with respect to secondary oven 15 °C. Temperature program for primary oven: 50 °C (1 min), increase to 320 °C (10 °C min⁻¹), 320 °C (3 min). +5 °C temperature offset on secondary column. Transfer line temperature was held at 250 °C. Mass detector was equipped with an electron ionization source and time-of-flight analyzer enabling unit mass resolution (scanned mass range was 30 – 500 *m/z*). The ion source chamber was held at 250 °C. ChromaTOF[®] v4.5 software (LECO Europe B.V.) was employed for instrument control and data processing. Selected compounds were identified by mass spectra library matching (NIST MS 2.2, USA). Compounds identified only in blanks (or that were highly abundant in blanks) were removed from analysis (e.g., silanes, siloxanes, propylphosphines etc.).

Protein digestion & nano-scale liquid chromatographic tandem mass spectrometry (nLC-MS/MS)

Urine proteins were precipitated with cold acetone and centrifuged (14,000 x *g*; 10 min; 0 °C), followed by re-suspension of dried pellets in digestion buffer (1% SDC, 100 mM TEAB; pH 8.5). Next, protein concentration in each lysate was determined (BCA assay kit; Fisher Scientific). We used tris(2-carboxyethyl)phosphine (TCEP; 5 mM; 60 °C; 60 min) as reducing agent and S-methyl methanethiosulfonate (MMTS; 10 mM; 10 min; RT) to block free cysteines. After trypsin digestion (1 µg per sample; 37 °C; overnight), peptides were desalted on a Michrom C18 column. We used reverse-phase nanocolumns (EASY-Spray column, 50 cm x 75 µm ID, PepMap C18, 2 µm particles, 100 Å pore size) for high resolution peptide separation. Eluting peptide cations were converted to gas-phase ions by electrospray ionization and analyzed on a Thermo Orbitrap Fusion (Q-OT-qIT; Thermo Fisher, Waltham, MA, USA) as previously described^{69,70}. LC-MS data were pre-processed with MaxQuant software (version 1.5.3.8)⁷¹. The false discovery rate (FDR) was set to 1% for both proteins and peptides. We specified a minimum peptide length of seven amino acids. The Andromeda search engine was used for MS/MS spectra search against Uniprot *Mus musculus* database (downloaded June 2015), containing 44,900 entries. From this database, all MUP and OBP sequences were removed and replaced by complete lists of MUPs (Ensembl) and OBPs⁷², respectively. We also added some sequences from TrEMBL that were missing in Uniprot (e.g., KLKs, BPIs, SPINKs, SCGB/ABPs, and LCNs). Enzyme specificity was set as C-terminal to Arg and Lys, also allowing cleavage at proline bonds⁷³ and a maximum of two missed cleavages. Quantifications were performed using label-free algorithms⁷⁴ with a combination of unique and razor peptides.

VNO slice preparation

For confocal Ca²⁺ imaging, acute coronal VNO slices were prepared as previously described^{38,61}. Briefly, C57BL/6 and BALB/c mice were euthanized by brief exposure to a CO₂ atmosphere, cervical dislocation and decapitation. The lower jaw and palate were rapidly removed. The VNO was dissected, embedded in 5% low-gelling temperature agarose (VWR International, Erlangen, Germany), placed in ice-cold oxygenated S₂, and coronal slices (150 µm) were cut on a VT1000S vibrating microtome (Leica Biosystems, Nussloch, Germany). For each experiment, we routinely prepared VNO slices along the organ's entire anterior-to-posterior axis (not including the most anterior tip, where the VNO lumen tapers into the vomeronasal duct, and

the most posterior part, the lumen “twists” toward the ventral aspect and its volume decreases ⁷⁵[75](#). This usually yields ~7 slices per individual experiment / session. Therefore, we routinely sample and average across the entire VNO anterior-to-posterior axis for each experiment. Slices were transferred to a submerged, chilled, and oxygenated storage chamber with circulating S_2 until use.

Ca²⁺ imaging

In vitro imaging of VSN activity in acute coronal VNO slices was performed as described ³⁸[38](#). Briefly, for bulk loading, slices were incubated (90 min; 5°C) in circulating S_2 with the Ca²⁺-sensitive dye CAL520/AM (4.5 μM; Biomol, Hamburg, Germany) and 0.05 % Pluronic® F-127 (20 % solution in DMSO; Thermo Fisher Scientific, Schwerte, Germany). After washing (5x, S_2), slices were transferred to a recording chamber (Luigs & Neumann, Ratingen, Germany) mounted on an upright fixed-stage scanning confocal microscope (TCS SP5 DM6000CFS, Leica Microsystems) equipped with a 20x / 1.0 NA water immersion objective (HCX APO L, Leica Microsystems), and infrared-optimized differential interference contrast optics. Slices were continuously superfused with oxygenated S_2 (~5 ml / min; gravity flow). CAL520 was excited at 488 nm (multi-line argon laser; <25% laser power) and fluorescence was detected within a 500–600 nm spectral band. Changes in cytosolic Ca²⁺ were monitored over time at 1.0 Hz frame rate (1024 x 512 pixels; 400 Hz bidirectional scanning frequency) using LAS AF software (Leica Microsystems).

Experimental Design and Statistical Analysis

Ca²⁺ imaging in VNO slices

All data were obtained from independent experiments performed in ≥3 sessions using ≥3 different animals. Individual numbers of cells / experiments (n) are denoted in figures and/or captions. Since, for each experiment, we prepare slices along the VNO’s anterior-to-posterior axis (see above), we routinely sample and average across the entire VNO anterior-to-posterior axis for each experiment. Data were analyzed offline using Leica LAS AF 2.4 (Leica Microsystems), ImageJ 1.51n (Wayne Rasband, National Institutes of Health, USA), MATLAB R2017b (MathWorks, Natick, MA), and Excel (Microsoft, Seattle, WA) software. If not stated otherwise, results are presented as box-and-whisker plots, where boxes represent the first-to-third quartiles and whiskers represent the 10th and 90th percentiles, respectively. Outliers (1.5 IQR) are plotted individually. The central band represents the population median ($P_{0.5}$). Statistical analyses were performed using unpaired t-tests, Wilcoxon signed-rank tests, Mann-Whitney U tests, and two-sample Kolmogorov-Smirnov tests (as dictated by data distribution and experimental design). Tests and corresponding *p*-values that report statistical significance (≤0.05) are individually specified in figure captions.

For individual cell analysis, regions-of-interest (ROIs) were defined to outline essentially all depolarization-sensitive (S_3) somata per field of view, based on DIC imaging of cell morphology at rest. After movement correction (StackReg/Rigid Body transformation plugin ⁷⁶[76](#) in ImageJ) of time-lapse image stacks, changes in relative fluorescence intensity were calculated as $\Delta F/F$ and measured in arbitrary units. Neurons were classified as ‘responsive’ when showing stimulus-dependent Ca²⁺ elevations in somata according to the following three criteria ³⁸[38](#), ⁷⁷[77](#): (i) exposure to high extracellular K⁺ concentrations (50 mM; S_3) induced a robust Ca²⁺ transient; (ii) for at least one stimulation with diluted urine, a transient increase in fluorescence intensity was observed during the stimulation period; and (iii) the signal peak intensity exceeded the average prestimulation baseline intensity plus two standard deviations for a continuous period of at least 3 s ($I_{resp} > I_{baseline} + 2 \times SD(I_{baseline})$). All responses were normalized to positive controls (i.e., responses evoked by elevated extracellular K⁺). Responsive neurons were categorized as: (i) neurons only sensitive to depolarization (K⁺); (ii) *specialist* neurons that selectively responded to one of two presented urine stimuli (and K⁺); and (iii) *generalist* neurons that responded to both urine stimuli (and K⁺). Raw data (i.e., original intensity *versus* time traces) from each responsive

cell were visually inspected to control for potential unspecific signals (e.g., caused by spontaneous activity). Based on peak amplitudes recorded from each urine-sensitive neuron (both generalists and specialists), we calculated a cell-specific *response index* (RI):

$$RI = \frac{\text{mean}\left(\frac{\Delta F}{F}\right)_{stim1} - \text{mean}\left(\frac{\Delta F}{F}\right)_{stim2}}{\text{mean}\left(\frac{\Delta F}{F}\right)_{stim1} + \text{mean}\left(\frac{\Delta F}{F}\right)_{stim2}}$$

where $\text{mean}\left(\frac{\Delta F}{F}\right)$ is the average signal amplitude evoked by consecutive stimulation with the same stimulus (i.e., either stim1 or stim2). As a measure of how reliably the same stimulus evokes a response upon consecutive exposures, we additionally calculated a cell-specific *Reliability Index* (Rel), which is based either on response amplitudes or their integrals (i.e., area under curve (AUC)):

$$Rel = \frac{\text{response}\left(\frac{\Delta F}{F} \text{ or } AUC\right)_{1st \text{ stim}} - \text{response}\left(\frac{\Delta F}{F} \text{ or } AUC\right)_{2nd \text{ stim}}}{\text{response}\left(\frac{\Delta F}{F} \text{ or } AUC\right)_{1st \text{ stim}} + \text{response}\left(\frac{\Delta F}{F} \text{ or } AUC\right)_{2nd \text{ stim}}}$$

where consecutive responses of similar strength (i.e., high reliability) are reflected by a small Rel value.

Proteomics and metabolomics

Pairwise comparison between different urine samples allowed comparative analysis of physiological activity (VSN Ca^{2+} signals) *versus* urine composition (molecular content). Chemical content was categorized as follows: (i) a specific compound is considered “present” in urine from a given sex/strain combination (i.e., male or female BALB/c, C57BL/6, or wild, respectively) if it is identified in ≥ 3 out of 10 individual samples; (ii) a compound is thus considered “absent” if it is found only in ≤ 2 individual samples. Accordingly, compounds that are detected as “present” in urine samples from both sex/strain combinations being compared are designated as *generic* in binary comparisons. By contrast, a compound that is “present” in only one of the two sex/strain combinations is designated as *specific*. To identify compounds that are *enriched* in a given sex/strain combination (**Figure 2a,b**) we raised the criterion for a “present” call to identification in ≥ 6 out of 10 individual samples.

To quantify concentration differences between samples we calculated *Concentration Indices* (CIs) for generic compounds, both VOCs (GCxGC-MS) and peptides/proteins (nLC-MS/MS):

$$CI = \frac{\text{mean conc. } (X_{group1}) - \text{mean conc. } (X_{group2})}{\text{mean conc. } (X_{group1}) + \text{mean conc. } (X_{group2})}$$

where *mean conc.* is the average concentration of a given compound *X* among all ten samples within a group (i.e., specific sex/strain combination).

Proteomic and metabolomic analysis

Because both GCxGC-MS and nLC-MS/MS data have similar (negative binomial) distributions, we processed them using the same procedure. First, we performed Sparse Partial Least Squares Analysis - sPLS-DA⁷⁸ to detect potential sources of variation in quantile normalized datasets. Next, for pairwise comparisons, data reduction eliminated all table entries (rows) if a given metabolite or protein was detected only in ≤ 3 individuals in both groups. However, if a compound was found in samples from ≥ 4 individuals in a group (e.g., in males or females), we included the full table entry (row). Accordingly, a chemical is considered “unique” / group-specific if found in ≥ 4 individual samples from one group, but is missing in all 10 samples of the other group. For calculation of sexual dimorphism, we used the Power Law Global Error Model - PLGEM⁷⁹ to detect differentially expressed / abundant proteins and VOCs. To detect the importance of

significant molecules in discriminating either sex or strain, we used a machine learning technique – Random Forest for Classification ⁴³[\[43\]](#) (implemented in R software) – and inferred feature importance scores (Gini importance ⁴³[\[43\]](#)) that provide relative rankings of individual variable relevance. nLC-MS/MS proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE ⁸⁰[\[80\]](#) partner repository with the dataset identifier PXD042324. Metabolomics data have been deposited to the EMBL-EBI MetaboLights database with the identifier MTBLS7439.

Supplemental information

Figure Supplements

- *Figure 1 – figure supplement 1*
- *Figure 2 – figure supplement 1*
- *Figure 2 – figure supplement 2*
- *Figure 2 – figure supplement 3*
- *Figure 5 – figure supplement 1*

Tables

- *Table 1 – relating to Figure 2*
- *Table 2 – relating to Figure 2 – figure supplement 1*

Supplementary Figures

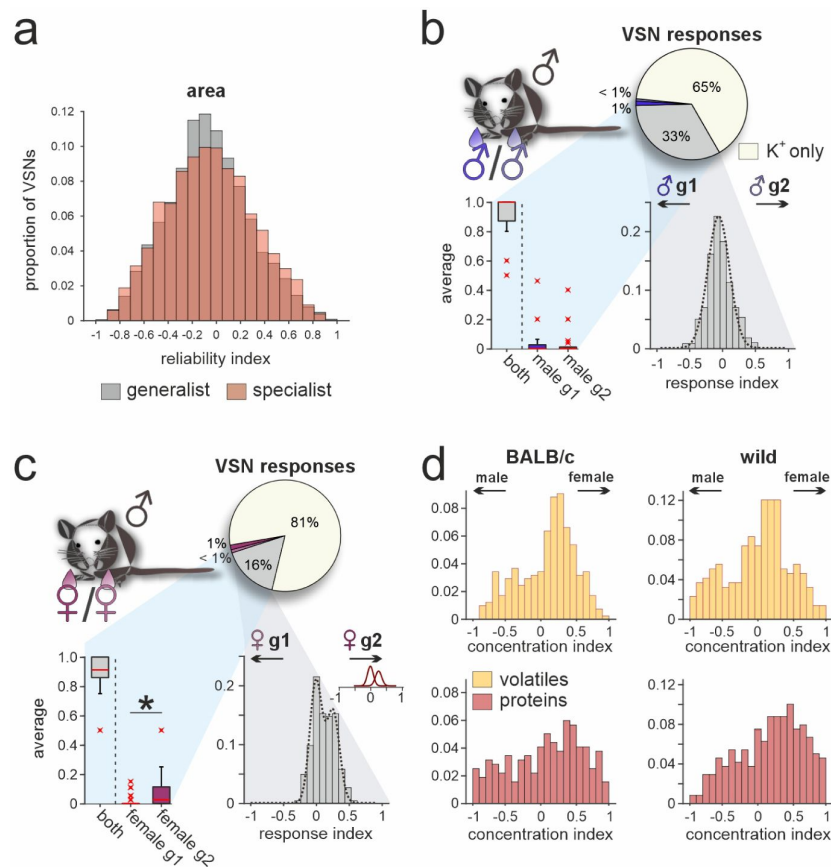


Figure 1 – figure supplement 1

(a) Area under curve response reliability index histograms of all generalist VSNs (gray; $n = 10,258$) and all specialist VSNs (red; $n = 6,457$) recorded from in this study. Note that for both response types, indices are normally distributed. While histograms peak around zero, distributions are much broader as compared to amplitude-based reliability index histograms (Fig. 1d). (b & c) Quantification of VSN responses recorded from male C57BL/6 mice challenged with urine stimuli from two different male C57BL/6 mouse cohorts (b) or two different female C57BL/6 mouse cohorts (c). Pie charts (top) illustrate the proportions of generalist (gray; 33 % (b) and 16 % (c), respectively) and specialist neurons ($\leq 1\%$; dark versus light blue (b) and dark versus light red (c), respectively) among all K⁺-sensitive VSNs ($n = 1867$ (b) and $n = 1946$ (c), respectively). Box- and-whisker plots (bottom, left) break down the urine-sensitive neurons by categories and illustrate the generalists-to-specialists ratios over individual experiments ($n = 21$). Boxes represent the first-to-third quartiles. Whiskers represent the 10th and 90th percentiles, respectively. Outliers (1.5 IQR; red x) are plotted individually. The central red band represents the population median ($P_{0.5}$). Response index histograms (bottom, left) depict signal amplitude strength distributions from generalist VSNs that responded to both paired stimuli (i.e., cohort / group 1 (g1) and cohort / group 2 (g2)). Multi-peak fitting with Gaussian curves (dashed lines) resulted either in a narrow single Gaussian that centers close to zero (b; peak = -0.06 , $\sigma = 0.16$) or two additive Gaussian curves that best represented the calculated index distribution (c; peak 1 = -0.01 , peak 2 = 0.25). Asterisk (*) indicates statistical significance, $p < 0.05$; Wilcoxon signed-rank test. (d) Concentration index histograms that outline individual compound concentration imbalances of VOCs (yellow; top) and proteins (red; bottom) that are found in both male and female urine samples of either BALB/c (left) or wild (right) mice. Histograms are heterogeneous and data are not normally distributed.

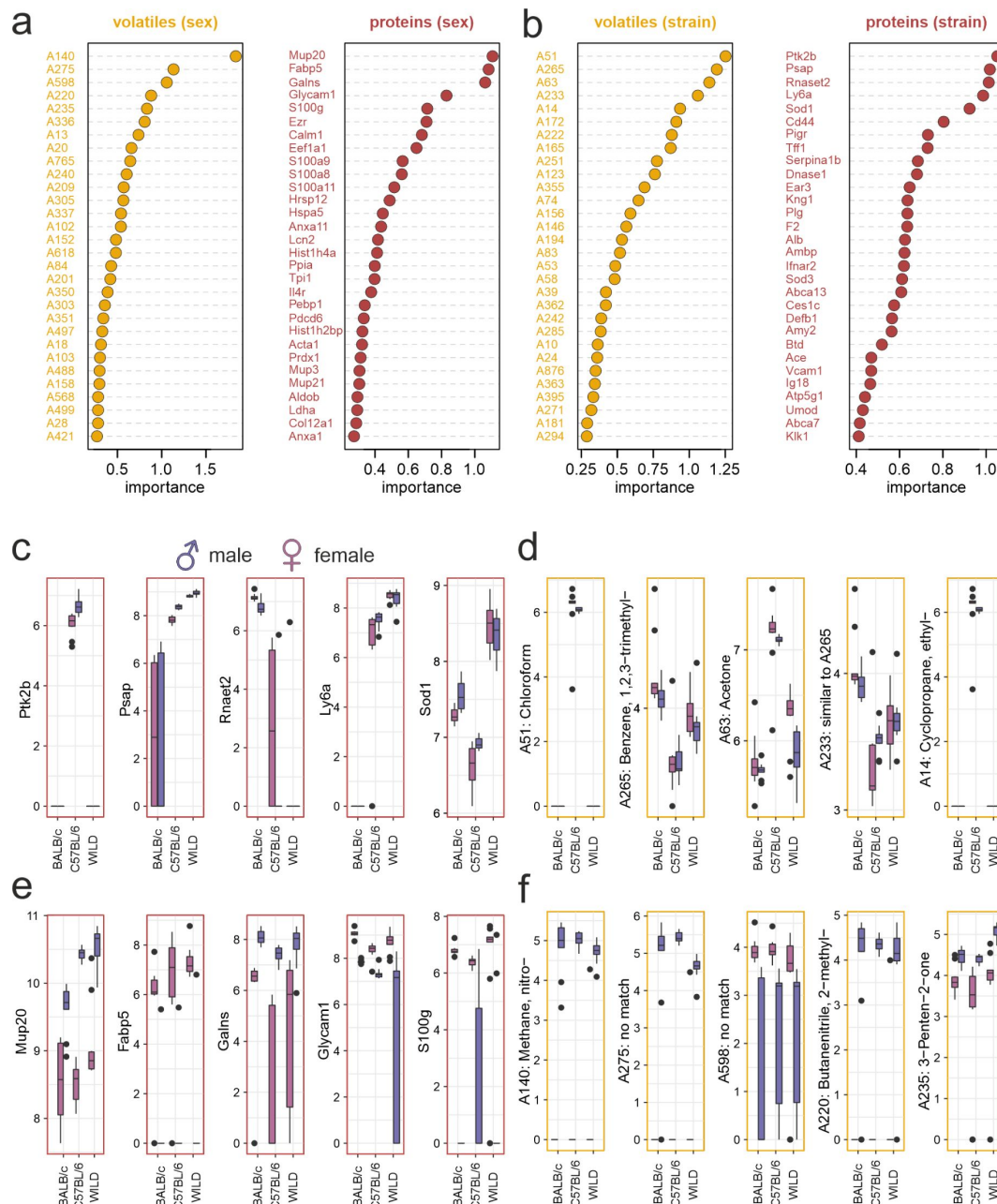


Figure 2 - figure supplement 1

(a&b) After training Random Forest classifiers⁴³, lists of “Gini importance”⁴³ scores rank urine VOCs (a; yellow) and proteins (b; red) that are likely most informative to decode sex and strain, respectively. (c - f) Box-and-whisker plots quantify concentrations of the respective five VOCs (c&e) and proteins (d&f) that are ranked most important / informative (a&b). Data are plotted as log2 signal intensity for each of the six sex/strain combinations (10 individual samples each). Red (female) and blue (male) boxes represent sex; strain as indicated. Boxes represent the first-to-third quartiles. Whiskers represent the 10th and 90th percentiles, respectively. Outliers (1.5 IQR; black dots) are plotted individually. Central bands represent the population median ($P_{0.5}$).

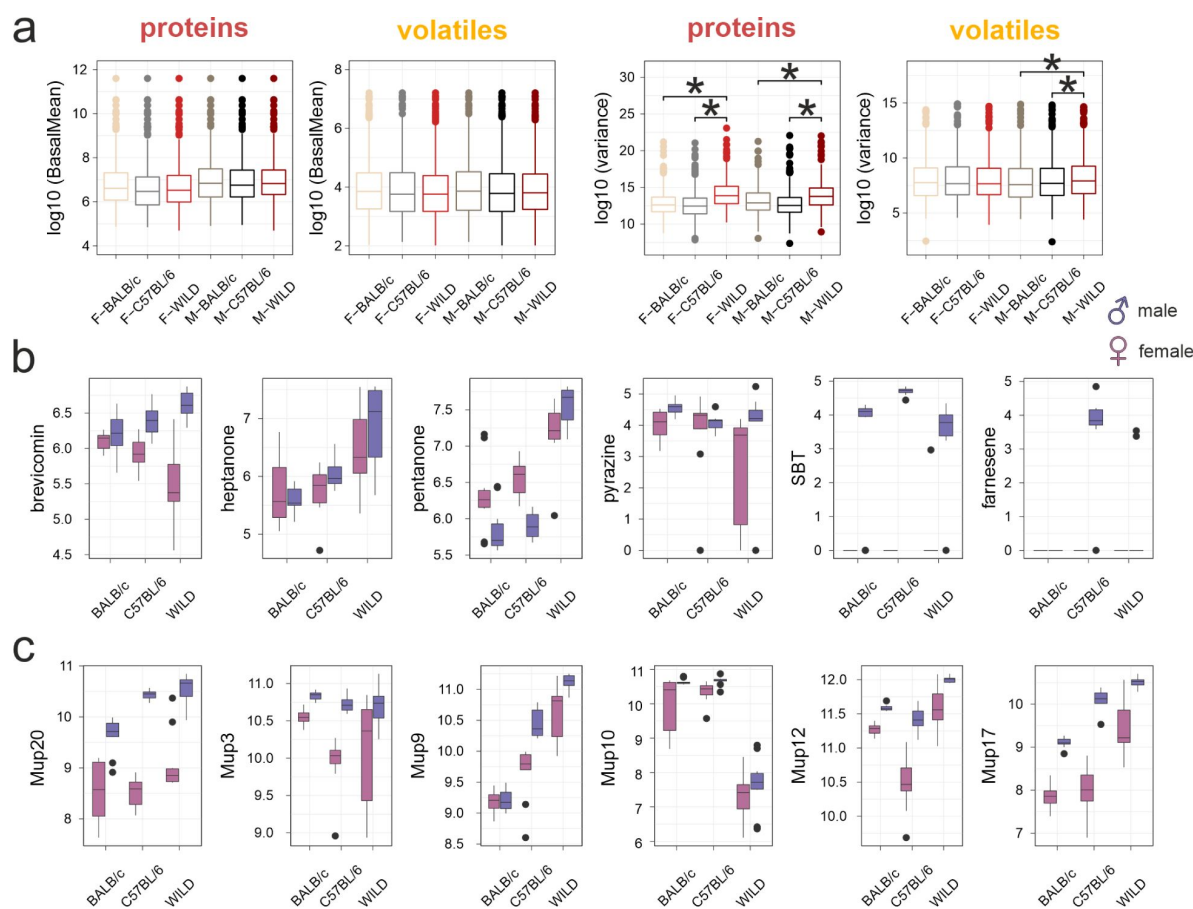


Figure 2 – figure supplement 2

(a) Box-and-whisker plots illustrating mean protein and VOC content (left) as well as individual compound concentration variance (right) for the six sex/strain combinations (10 individual samples each). Asterisks (*) indicate statistical significance, $p < 0.05$, Kruskal-Wallis test. **(b&c)** Box-and-whisker plots quantify concentrations of previously reported VOC **(b)** and Mup **(c)** semiochemicals, respectively. Data are plotted as log2 signal intensity for each of the six sex/strain combinations (10 individual samples each). Red (female) and blue (male) boxes represent sex; strain as indicated. Boxes depict the first-to-third quartiles. Whiskers represent the 10th and 90th percentiles, respectively. Outliers (1.5 IQR; black dots) are plotted individually. Central bands represent the population median ($P_{0.5}$).

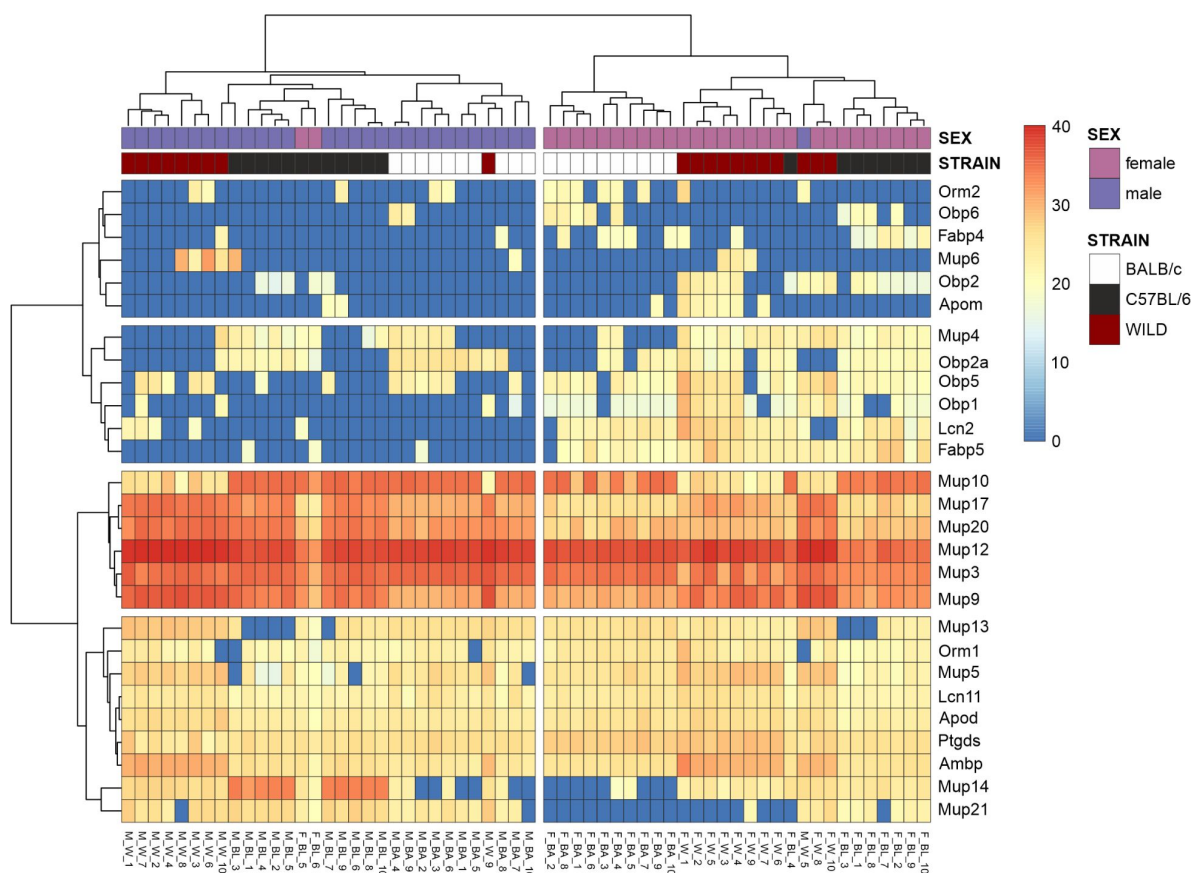


Figure 2 – figure supplement 3

Hierarchical clustering of mouse urine lipocalin content reveals an unexpectedly diverse repertoire of 27 lipocalins. The horizontal tree clusters lipocalins according to proteomic homology based on similarities in protein abundances. The vertical tree clusters the 60 individual urine samples (10 samples for each of the 6 sex/strain combinations) according to patterns of lipocalin abundance (pseudocolors).

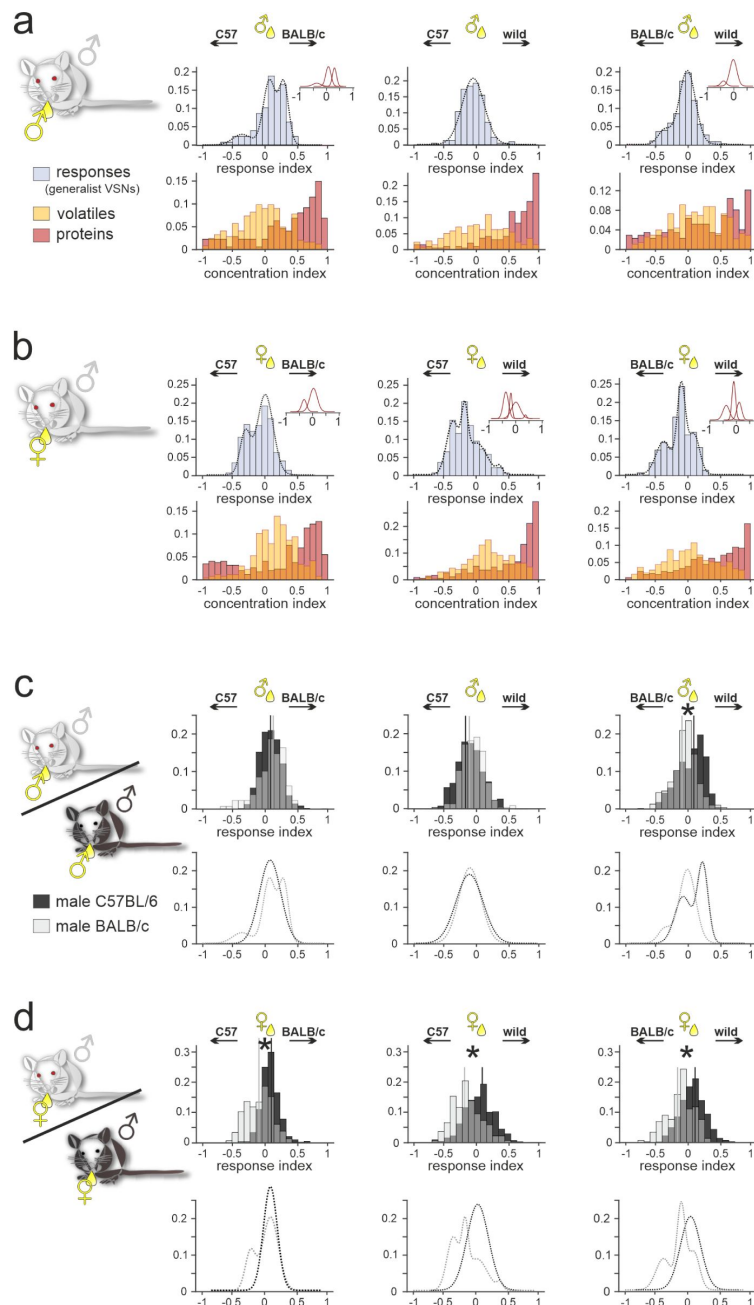


Figure 5 – figure supplement 1

(a&b) Comparison of male BALB/c generalist VSN response preferences, upon exposure to paired male **(a)** and female **(b)** urine stimuli from different strains, with strain-dependent concentration imbalances among VOCs and proteins, respectively. Response index histograms (top rows) depict distributions of generalist data outlined in **Figure 5** (gray bars). With one exception (**a** (middle), male C57BL/6 versus male wild), histograms are not well fitted by single Gaussian curves. Rather, histogram shape is best represented by additive multi-peak fitting (dashed lines) with two (**a** (right), **b** (left)), three (**a** (left), **b** (right)), or even four (**b** (middle)) Gaussians. Individual Gaussian curves center at -0.37 , 0.07 , and 0.3 (**a**; left), -0.06 (**a**; middle), -0.39 and -0.02 (**a**; right), -0.32 and 0.04 (**b**; left), -0.37 , -0.18 , 0.01 , and 0.37 (**b**; middle), -0.38 , -0.1 , and 0.11 (**b**; right), respectively. For comparison, transparent and overlapping concentration index histograms (bottom rows) for VOCs (yellow) and proteins (red) are also shown (data correspond to results shown in **Figure 4**). **(c&d)** Comparative overlay of generalist VSN response index histograms (top rows) and corresponding fits (bottom rows) calculated for recordings from male BALB/c (white transparent bars) versus male C57BL/6 (black bars) animals that were stimulated with male **(c)** and female **(d)** urine, respectively. Vertical lines represent mean values. Asterisks (*) indicate statistical significance between BALB/c and C57BL/6 response index distributions, $p < 0.05$, two-sample Kolmogorov-Smirnov test.

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Reviewer #1 (Public Review):

In this manuscript, Nagel et al. sought to characterize the composition of urinary compounds, some of which are putative chemosignals. They used urines from adult males and females in three different strains, including one wild-derived strain. By performing mass spectrometry of two classes of compounds: volatile organic compounds and proteins, they found that urines from inbred strains are qualitatively similar to those of a wild strain. This finding is significant because there is a high degree of diversity in different inbred strains and wild mice, with respect to the polymorphisms of chemosensory receptor genes and expression of vomeronasal ligands previously identified. Notably, their study did not characterize steroids, which represent a major class of urinary chemosignals activating vomeronasal neurons. Therefore, important future studies should address the strain dependence of steroid composition in urines.

In the second part of this work, the authors used calcium imaging to monitor the pattern of vomeronasal neuron responses to these urines. By performing pairwise comparisons, the authors found a large degree of strain-specific response and a relatively minor response to sex-specific urinary stimuli. This is a finding generally in agreement with previous calcium imaging work by Ron Yu and colleagues in 2008. The authors extend the previous work by using urines from wild mice. They further report that the concentration diversity of urinary compounds in different urine batches is largely uncorrelated with the activity profiles of these urines. In addition, the authors found that the patterns of vomeronasal neuron response to urinary cues are not identical when measured using different recipient strains.

The pitfalls of this study are the omission of steroids for the mass spectrometry experiments and the indirect (correlational) nature of their mass spectrometry data and activity data. Whether the urinary compounds identified in this study activate vomeronasal neurons were not tested.

Nevertheless, the major contribution of this work is the identification of specific molecules in mouse urines. This work is likely to be of significant interest to researchers in chemosensory signaling in mammals and could provide a systematic avenue to exhaustively identify additional pheromones in mice.

<https://doi.org/10.7554/eLife.90529.2.sa2>

Reviewer #2 (Public Review):

This manuscript by Nagel et al provides a comprehensive examination of the chemical composition of mouse urine (an important source of semiochemicals) across strain and sex, and correlates these differences with functional responses of vomeronasal sensory neurons (an important sensory population for detecting chemical social cues). The strength of the work lies in the careful and comprehensive imaging and chemical analyses, the rigor of quantification of functional responses, and the insight into the relevance of olfactory work on lab-derived vs wild-derived mice.

With regards to the chemical analysis, the reader should keep in mind (and the authors acknowledge) that a difference in the concentration of a chemical across strain or sex does not necessarily mean that that chemical is used for chemical communication. In the most extreme case, the animals may be completely insensitive to the chemical. Thus, the fact that the repertoire of proteins and volatiles could potentially allow sex and/or strain discrimination, it is unclear to what degree both are used in different situations.

<https://doi.org/10.7554/eLife.90529.2.sa1>

Reviewer #3 (Public Review):**Summary:**

The manuscript by Nagel, et al. describes studies of mouse vomeronasal sensory neuron (VSN) tuning to mouse urine samples across different sexes and strains, including wild mice, alongside mass spectrometry analysis of the same samples. The authors performed live Ca²⁺ imaging (CAL520 dye) of VSNs in acute vomeronasal organ (VNO) slices to determine how VSNs are tuned to pairs of stimuli that differ in their origin (e.g. male C57BL/6 versus male BALB/c urine, male C57BL/6 versus female C57BL/6, etc.). For each pair of tested odorants, the results measure the proportion of VSNs that respond to both stimuli ("generalists") or just one of the two ("specialists"), as well as metrics of tuning preference and response reliability. The authors find in most cases that generalists make up a larger proportion of responsive VSNs than specialists, but several pairwise comparisons showed a high degree of strain selectivity. Notably, the authors evaluated VSN tuning in both male C57BL/6 and male BALB/c VNOs, finding strain-dependent differences in the representation of mouse urine. Alongside these measurements of VSN tuning, the authors report results of mass spectrometry analyses of volatiles and proteins in the same urine samples. These analyses indicated a number of molecules in each category that vary across sex and strain, and therefore represent candidate vomeronasal ligands. However, this study did not directly test whether any of these candidate molecules drives VSN activity. Overall, this work provides solid information related to mouse vomeronasal chemosensation.

Strengths:

A strength of the current study is its focus on characterizing the neural responses of the VNO to urine derived from wild mice. The majority of existing vomeronasal system research has relied on the use of inbred strains for both neural response recordings and investigations of candidate vomeronasal system ligands. Inbreeding in laboratory environments may alter the chemical composition of bodily secretions, thereby potentially changing the information they contain. Moreover, the more homogeneous nature of inbred strains could be critical when studying the AOS mediated social aspects. If there exist noticeable differences in the chemical composition of secretions from wild animals compared to inbred strains, this would suggest that future research must consider natural sources of candidate ligands outside of inbred strains. This work identifies some intriguing differences, worthy of further exploration, between the urine composition of wild mice versus inbred mice, as well as disparities in how the VNO responds to urine from these different sources. However, the molecular composition and VNO responsiveness to wild mouse urine was found to be highly overlapping with inbred mouse urine, supporting the continued investigation of candidate ligands found in inbred mouse urine.

Another positive aspect of this work is its use of the same set of stimuli as a previous study by the same authors (Bansal et al., 2021) in the downstream accessory olfactory bulb. The consistency in stimulus selection facilitates a comparison of information processing of sex and strain information from the sensory periphery to the brain. Although comparisons between the two connected regions are not a focus of this work, and methodological differences (e.g., Ca²⁺ imaging versus electrophysiology) may introduce caveats into comparisons, the support of "apples to apples" comparisons across connected circuits is critical to progress in the field.

Finally, this study directly measured VSN tuning in both male C57BL/6 and male BALB/c VNOs, finding subtle but important differences in the representation of mouse urine in these two recipient strains. Given that there is a long history of behavioral research into strain-specific differences in social behavior, this research paves the way for future studies into how different mouse strains detect and process social chemosignals.

Weaknesses:

One of the primary objectives in this study is to ascertain the extent to which the response profiles of VSNs are specific to sex and strain. The design of these Ca²⁺ imaging experiments uses a simple stimulus design, using two interleaved bouts of stimulation with pairs of urine (e.g., male versus female C57BL/6, male C57BL/6 versus male BALB/c) at a single dilution factor (1:100). This introduces two significant limitations: (1) the "generalist" versus "specialist" descriptors pertain only to the specific pairwise comparisons made and (2) there is no information about the sensitivity/concentration-dependence of the responses.

The functional measurements of VSN tuning to various pairs of urine stimuli are presented alongside mass spectrometry-based comparisons. However, the mass spectrometry-based analysis was performed separately from VSN tuning experiments/analysis. The juxtaposition of these measurements may give some readers the impression that VSN tuning measurements were integrated with molecular profiling (i.e., that the molecular diversity was causally related physiological responses). This is a hypothesis raised by the parallel studies, but not a supported conclusion of the current work.

The impact of mass spectrometry findings is acknowledged to be limited to nonvolatile organic compounds and proteins/peptides, and that it is possible that few of these candidate molecules are active in the VNO. Moreover, it remains possible that the VSN responses are driven mostly by small nonvolatiles (e.g., polar steroids), a class of strong VSN ligands that were excluded from molecular analysis.

<https://doi.org/10.7554/eLife.90529.2.sa0>

Author response:

The following is the authors' response to the original reviews.

General comments

All three experts have raised excellent ideas and made important suggestions to extend the scope of our study and provide additional information. While we fully acknowledge that these points are valid and would provide exciting new knowledge, we also should not lose track of the fact that a single study cannot cover all bases. Sulfated steroids, for example, are clearly essential components of mouse urine. Unfortunately, however, all chemical analysis approaches are limited and the one we opted for is not suitable for analysis of such signaling molecules. Future studies should certainly focus on these aspects. The same holds true for the fact that we do not know which of the identified compounds are actually VSN ligands. These are inherent limitations of the approach, and we are not claiming otherwise.

Reviewer #1 (Public Review):

(1) In this manuscript, Nagel et al. sought to comprehensively characterize the composition of urinary compounds, some of which are putative chemosignals. They used urines from adult males and females in three different strains, including one wild-derived strain. By performing mass spectrometry of two classes of compounds: volatile organic compounds and proteins, they found that urines from inbred strains are qualitatively similar to those of a wild strain. This finding is significant because there is a high degree of genetic diversity in wild mice, with chemosensory receptor genes harboring many polymorphisms.

We agree and thank the Reviewer for his / her positive assessment.

(2) In the second part of this work, the authors used calcium imaging to monitor the pattern of vomeronasal neuron responses to these urines. By performing pairwise comparisons, the authors found a large degree of strain-specific response and a relatively minor response to sex-specific urinary stimuli. This is a finding generally in agreement with previous calcium imaging work by Ron Yu and colleagues in 2008. The authors extend the previous work by using urines from wild mice. They further report that the concentration diversity of urinary compounds in different urine batches is largely uncorrelated with the activity profiles of these urines. In addition, the authors found that the patterns of vomeronasal neuron response to urinary cues are not identical when measured using different recipient strains. This fascinating finding, however, requires an additional control to exclude the possibility that this is not due to sampling error.

We thank Reviewer 1 for pointing this out. We agree that this is truly a “fascinating finding.” Reviewer 1 emphasizes that we need to add an “additional control to exclude [...] that this is not due to sampling error”, and he / she elaborates on the required control in his / her Recommendations For The Authors (see below). Reviewer 1 states that “for Fig. 5, in order to conclude that the same urine activates a different population of VSNs in two different strains, a critical control is needed to demonstrate that this is not due to the sampling variability - as compositions of V1Rs and V2Rs could vary between different slices, one preferred control is to use VNO slices from the same strain and compare the selectivity used here across the A-P axis.” Importantly, we believe that this is already controlled for. In fact, for each experiment, we routinely prepare VNO slices along the organ’s entire anterior-to-posterior axis (not including the most anterior tip, where the VNO lumen tapers into the vomeronasal duct, and the most posterior part, the lumen “twists” toward the ventral aspect and its volume decreases (see Figs. 7 & S7 in Hamacher et al., 2024, Current Biology)). This usually yields ~7 slices per individual experiment / session. Therefore, we routinely sample and average across the entire VNO anterior-to-posterior axis for each experiment. In Fig. 5, in which we analyzed whether the “same urine activates a different population of VSNs in two different strains”, individual independent experiments from each strain (C57BL/6 versus BALB/c) amounted to (a) n = 6 versus n = 8; (b) n = 10 versus n = 10; (c) n = 7 versus n = 9; (d) n = 9 versus n = 10; (e) n = 10 versus n = 9; and (f) n = 12 versus n = 10. Together, we conclude that it is very unlikely that the considerably different response profiles measured in different recipient strains result from a “sampling error.”

To clarify this point in the revised manuscript, we now explain our sampling routine in more detail in the Materials and Methods. Moreover, we now also refer to this point in the Results.

(3) There are several weaknesses in this manuscript, including the lack of analysis of the compositions of sulfated steroids and other steroids, which have been proposed to be the major constituents of vomeronasal ligands in urines and the indirect (correlational) nature of their mass spectrometry data and activity data.

Reviewer 1 is correct to point out that our chemical profiling approach omits (sulfated) steroids. We are aware of this weakness. We deliberately decided to omit steroids as well as other nonvolatile small organic molecules for three main reasons: (i) as the reviewer points out, (sulfated) steroid composition has been the focus of analysis in several previous studies and there is ample published information available on their role as VSN stimuli; (ii) the analytical tools available to us do not allow comprehensive profiling of non-volatile small organic molecules; employing two-dimensional head-space GC-MS as well as LC-MS/MS is not suitable for steroid detection; and (iii) the relatively small sample volumes forced us to prioritize and focus on specific chemical classes (in our case, VOCs and proteins). We made an effort to use of the exact same stimuli as previously employed to investigate sensory

representations in the accessory olfactory bulb (AOB) (Bansal et al., 2021), a feature that we consider a strength of the current study. However, this entailed that we had to effectively split our samples, further reducing the available sample volume.

We acknowledge that we did not sufficiently describe our rationale for focusing on VOCs and proteins on the previous version of the manuscript (nor did we discuss the known role of (sulfated) steroids in VSN signaling in adequate detail). We have now made an effort to address these shortcomings in the revised manuscript. Specifically, we have added new text to the Introduction (“Prominent molecularly identified VSN stimuli include various sulfated steroids (Celsi et al., 2012; Fu et al., 2015; Haga-Yamanaka et al., 2015, 2014; Isogai et al., 2011; Nodari et al., 2008; Turaga and Holy, 2012), which could reflect the dynamic endocrine state of an individual.”) and the Discussion (“Notably, our chemical profiling approach omits (sulfated) steroids other non-volatile small organic molecules, which have previously been identified in mouse urine as VSN stimuli (Nodari et al., 2008). Caution should thus be exerted to not attempt to fully explain VSN response specificity based on VOC and protein content alone.” & “In line with the notion of highly selective vomeronasal sampling is our observation that the concentration differences between compounds shared among strains, which are often substantial, are not reflected by similarly pronounced differences in response strength among generalist VSNs. There are several, not necessarily mutually exclusive explanations for this finding: First, concentration could simply not be a read-out parameter for VSNs, which would support previous ideas of concentration-invariant VSN activity (Leinders-Zufall et al., 2000). Second, the concentrations in freshly released urine could just exceed the dynamic tuning range of VSNs since, particularly for VOCs, natural signals (e.g., in scent marks) must be accessible to a recipient for a prolonged amount of time (sometimes days). A similar rationale could explain the increased protein concentrations in male urine, since male mice use scent marking to establish and maintain their territories and urinary lipocalins serve as long-lasting reservoirs of VOCs (Hurst et al., 1998). Third, generalist VSNs might sample information only from a select subset of urinary compounds, which, given their role as biologically relevant chemosignals, might be released at tightly controlled (and thus similar) concentrations. In fact, in the most extreme scenario, several compounds that do display substantial strain- and/or sex-specific differences in concentration might not act as chemosignals at all. Forth, to some extent, different response profiles could be attributed to non-volatile small organic molecules such as steroids (Nodari et al., 2008), which were beyond the focus of our chemical analysis.”).

(4) Overall, the major contribution of this work is the identification of specific molecules in mouse urines. This work is likely to be of significant interest to researchers in chemosensory signaling in mammals and provides a systematic avenue to exhaustively identify vomeronasal ligands in the future.

We thank the Reviewer for his / her generally positive assessment.

Reviewer #2 (Public Review):

(1) This manuscript by Nagel et al provides a comprehensive examination of the chemical composition of mouse urine (an important source of semiochemicals) across strain and sex, and correlates these differences with functional responses of vomeronasal sensory neurons (an important sensory population for detecting chemical social cues). The strength of the work lies in the careful and comprehensive imaging and chemical analyses, the rigor of quantification of functional responses, and the insight into the relevance of olfactory work on lab-derived vs wild-derived mice.

We thank the Reviewer for his / her generally positive assessment.

(2) With regards to the chemical analysis, the reader should keep in mind that a difference in the concentration of a chemical across strain or sex does not necessarily mean that that chemical is used for chemical communication. In the most extreme case, the animals may be completely insensitive to the chemical. Thus, the fact that the repertoire of proteins and volatiles could potentially allow sex and/or strain discrimination, it is unclear to what degree both are used in different situations.

Reviewer 2 is correct to point out that sex- and/or strain-dependent differences in urine molecular composition do not automatically attribute a signaling function to those molecules. We concur and, in fact, stress this point many times throughout the manuscript. In the Results, for example, we point out (i) that “in female urine, BALB/c-specific proteins are substantially underrepresented, a fact not reflected by VSN response profiles”, (ii) that “as observed in C57BL/6 neurons, the skewed distributions of protein concentration indices were not reflected by BALB/c generalist VSN profiles”, and (iii) that “VSN population response profiles do not reflect the global molecular content of urine, suggesting that the VNO functions as a rather selective molecular detector.” Moreover, in the Discussion, we state (i) that “caution should thus be exerted to not attempt to fully explain VSN response specificity based on VOC and protein content alone”; (ii) that, for several sex- and/or strain-specific molecules, none “has previously been attributed a chemosensory function. Challenging the mouse VNO with purified recombinant protein(s) will help elucidate whether such functions exist”; (iii) that “generalist VSNs might sample information only from a select subset of urinary compounds, which, given their role as biologically relevant chemosignals, might be released at tightly controlled (and thus similar) concentrations”; and (iv) that “to some extent, different response profiles could be attributed to non-volatile small organic molecules such as steroids (Nodari et al., 2008), which were beyond the focus of our chemical analysis.”

In the revised manuscript, we now aim to even more strongly emphasize the point made by Reviewer 2. In the Discussion, we have deleted a sentence that read: “Sex- and strain-specific chemical profiles give rise to unique VSN activity patterns.” Moreover, we have added the following statement: “In fact, in the most extreme scenario, several compounds that do display substantial strain- and/or sex-specific differences in concentration might not act as chemosignals at all.”

Reviewer #3 (Public Review):

(1) One of the primary objectives in this study is to ascertain the extent to which the response profiles of VSNs are specific to sex and strain. The design of these Ca²⁺ imaging experiments uses a simple stimulus design, using two interleaved bouts of stimulation with pairs of urine (e.g. male versus female C57BL/6, male C57BL/6 versus male BALB/c) at a single dilution factor (1:100). This introduces two significant limitations: (1) the “generalist” versus “specialist” descriptors pertain only to the specific pairwise comparisons made and (2) there is no information about the sensitivity/concentration-dependence of the responses.

Reviewer 3 points to two limitations of our VSN activity assay. He / she is correct to mention that characterizing a VSN as generalist or specialist based on a “pairwise comparison” should not be the basis of attributing such a “generalist” or “specialist” label in general (i.e., regarding the global stimulus space). We acknowledge this point, but we do not regard this as a limitation of our study since we are not investigating rather broad (i.e., multidimensional) questions of selectivity. All we are asking in the context of this study is whether VSNs - when being challenged with pairs of sex- or strain-specific urine samples - act as rather selective semiochemical detectors. Of course, one can always think of a study design that provides more information. However, we here opted for an assay that - in our hands - is robust, “low noise” (i.e., displays low intrinsic signal variability as evident from reliability index

calculations), ensures recovery from VSN adaptation (Wong et al., 2018), and, importantly, answers the specific question we are asking.

Regarding the second point (“there is no information about the sensitivity/concentrationdependence of the responses”), we would like to emphasize that this was not a focus of our study either. In fact, concentration-dependence of VSN activity has been a major focus of several previous studies referenced in our manuscript (e.g., Leinders-Zufall et al., 2000; He et al., 2008), albeit with contradictory results. In our study, we ask whether a pair of stimuli that we have shown to display, in part, strikingly different chemical composition (both absolute and relative) preferentially activates the same or different VSNs. With this question in mind, we believe that our assay (and its results) are highly informative.

(2) The functional measurements of VSN tuning to various pairs of urine stimuli are consistently presented alongside mass spectrometry-based comparisons. Although it is clear from the manuscript text that the mass spectrometry-based analysis was separated from the VSN tuning experiments/analysis, the juxtaposition of VSN tuning measurements with independent molecular diversity measurements gives the appearance to readers that these experiments were integrated (i.e., that the diversity of ligands was underlying the diversity of physiological responses). This is a hypothesis raised by the parallel studies, not a supported conclusion of the work. This data presentation style risks confusing readers.

As Reviewer 3 points out correctly “it is clear from the manuscript text that the mass spectrometry-based analysis was separated from the VSN tuning experiments/analysis.” In the figures, we try make the distinction between VSN response statistics and chemical profiling more obvious by gray shadows that link the plots depicting VSN response characteristics to the general pie charts.

We now also made an extra effort to avoid “confusing readers” by stating in the Discussion (i) that “caution should thus be exerted to not attempt to fully explain VSN response specificity based on VOC and protein content alone”; (ii) that, for several sex- and/or strain-specific molecules, none “has previously been attributed a chemosensory function. Challenging the mouse VNO with purified recombinant protein(s) will help elucidate whether such functions exist”; (iii) that “generalist VSNs might sample information only from a select subset of urinary compounds, which, given their role as biologically relevant chemosignals, might be released at tightly controlled (and thus similar) concentrations”; and (iv) that “to some extent, different response profiles could be attributed to non-volatile small organic molecules such as steroids (Nodari et al., 2008), which were beyond the focus of our chemical analysis.” Moreover, we have deleted a sentence that read: “sex- and strain-specific chemical profiles give rise to unique VSN activity patterns”, and we have added the following statement: “In fact, in the most extreme scenario, several compounds that do display substantial strain- and/or sex-specific differences in concentration might not act as chemosignals at all.”

However, we believe that there is value in presenting “VSN tuning measurements” next to “independent molecular diversity measurements.” While these are independent measurements, their similarity or, quite frequently, lack thereof are informative. We are sure that by taking the above “precautions” we have now mitigated the risk of “confusing readers.”

(3) The impact of mass spectrometry findings is limited by the fact that none of these molecules (in bulk, fractions, or monomolecular candidate ligands) were tested on VSNs. It is possible that only a very small number of these ligands activate the VNO. The list of variably expressed proteins - especially several proteins that are preferentially found in female urine - is compelling, but, again, there is no evidence presented that indicates whether or not these candidate ligands drive VSN activity. It is noteworthy that the

largest class of known natural ligands for VSNs are small nonvolatiles that are found at high levels in mouse urine. These molecules were almost certainly involved in driving VSN activity in the physiology assays (both "generalist" and "specialist"), but they are absent from the molecular analysis.

Reviewer 3 is right, of course, that at this point we have not tested the identified molecules on VSNs. This is clearly beyond the scope of the present study. We believe that the data we present will be the basis of (several full-length) future studies that aim to identify specific ligands and - best case scenario - receptor-ligand pairs. We find it hard to concur that our study, which provides the necessary basis for those future endeavors, is regarded as "incomplete". By design, all studies are somewhat incomplete, i.e., there are always remaining questions and we are not contesting that.

It is true, of course, that a class of "known natural ligands for VSNs are small nonvolatiles." As we replied above, our chemical profiling approach omits (sulfated) steroids. We are aware of this weakness. We deliberately decided to omit steroids as well as other non-volatile small organic molecules for three main reasons: (i) steroid composition has been the focus of analysis in several previous studies and there is ample published information available on their role as VSN stimuli; (ii) the analytical tools available to us do not allow comprehensive profiling of non-volatile small organic molecules; employing two-dimensional head-space GC-MS as well as LC-MS/MS is not suitable for steroid detection; and (iii) the relatively small sample volumes forced us to prioritize and focus on specific chemical classes (in our case, VOCs and proteins). We made an effort to use of the exact same stimuli as previously employed to investigate sensory representations in the accessory olfactory bulb (AOB) (Bansal et al., 2021), a fact that we consider a key strength of our current study. However, this entailed that we had to effectively split our samples, further reducing the available sample volume.

We acknowledge that we did not sufficiently describe our rationale for focusing on VOCs and proteins on the previous version of the manuscript (nor did we discuss the known role of (sulfated) steroids in VSN signaling in adequate detail). We have now made an effort to address these shortcomings in the revised manuscript. Specifically, we have added new text to the Introduction ("Prominent molecularly identified VSN stimuli include various sulfated steroids (Celsi et al., 2012; Fu et al., 2015; Haga-Yamanaka et al., 2015, 2014; Isogai et al., 2011; Nodari et al., 2008; Turaga and Holy, 2012), which could reflect the dynamic endocrine state of an individual.") and the Discussion ("Notably, our chemical profiling approach omits (sulfated) steroids other non-volatile small organic molecules, which have previously been identified in mouse urine as VSN stimuli (Nodari et al., 2008). Caution should thus be exerted to not attempt to fully explain VSN response specificity based on VOC and protein content alone." & "In line with the notion of highly selective vomeronasal sampling is our observation that the concentration differences between compounds shared among strains, which are often substantial, are not reflected by similarly pronounced differences in response strength among generalist VSNs. There are several, not necessarily mutually exclusive explanations for this finding: First, concentration could simply not be a read-out parameter for VSNs, which would support previous ideas of concentration-invariant VSN activity (Leinders-Zufall et al., 2000). Second, the concentrations in freshly released urine could just exceed the dynamic tuning range of VSNs since, particularly for VOCs, natural signals (e.g., in scent marks) must be accessible to a recipient for a prolonged amount of time (sometimes days). A similar rationale could explain the increased protein concentrations in male urine, since male mice use scent marking to establish and maintain their territories and urinary lipocalins serve as long-lasting reservoirs of VOCs (Hurst et al., 1998). Third, generalist VSNs might sample information only from a select subset of urinary compounds, which, given their role as biologically relevant chemosignals, might be released at tightly controlled (and thus similar) concentrations. In fact, in the most extreme scenario, several compounds that do display substantial strain- and/or sex-specific differences in concentration might not act as

chemosignals at all. Forth, to some extent, different response profiles could be attributed to non-volatile small organic molecules such as steroids (Nodari et al., 2008), which were beyond the focus of our chemical analysis.”).

Reviewer #1 (Recommendations For The Authors):

(1) I find that the study is highly valuable for researchers in this field. With the finding that wild mouse urines do not elicit significantly more variable responses from urines from inbred strains, researchers can now be reassured to use inbred strains to gain general insights on pheromone signaling.

A major omission of this study is non-volatile small organic molecules such as steroids. These compounds are the only molecular class in urine that have been identified to stimulate specific vomeronasal receptors to date. It is unclear to me that the specificity of VOC and proteins can alone fully explain the response specificity of the VSNs that have been monitored in this study. The discussion of this topic is highly beneficial for the readers.

Reviewer 1 is correct to point out that our chemical profiling approach omits (sulfated) steroids. We are aware of this weakness. We deliberately decided to omit steroids as well as other nonvolatile small organic molecules for three main reasons: (i) as the reviewer points out, (sulfated) steroid composition has been the focus of analysis in several previous studies and there is ample published information available on their role as VSN stimuli; (ii) the analytical tools available to us do not allow comprehensive profiling of non-volatile small organic molecules; employing two-dimensional head-space GC-MS as well as LC-MS/MS is not suitable for steroid detection; and (iii) the relatively small sample volumes forced us to prioritize and focus on specific chemical classes (in our case, VOCs and proteins). We made an effort to use of the exact same stimuli as previously employed to investigate sensory representations in the accessory olfactory bulb (AOB) (Bansal et al., 2021), a fact that we consider a key strength of our current study. However, this entailed that we had to effectively split our samples, further reducing the available sample volume.

We acknowledge that we did not sufficiently describe our rationale for focusing on VOCs and proteins on the previous version of the manuscript (nor did we discuss the known role of (sulfated) steroids in VSN signaling in adequate detail). We have now made an effort to address these shortcomings in the revised manuscript. Specifically, we have added new text to the Introduction (“Prominent molecularly identified VSN stimuli include various sulfated steroids (Celsi et al., 2012; Fu et al., 2015; Haga-Yamanaka et al., 2015, 2014; Isogai et al., 2011; Nodari et al., 2008; Turaga and Holy, 2012), which could reflect the dynamic endocrine state of an individual.”) and the Discussion (“Notably, our chemical profiling approach omits (sulfated) steroids other non-volatile small organic molecules, which have previously been identified in mouse urine as VSN stimuli (Nodari et al., 2008). Caution should thus be exerted to not attempt to fully explain VSN response specificity based on VOC and protein content alone.” & “In line with the notion of highly selective vomeronasal sampling is our observation that the concentration differences between compounds shared among strains, which are often substantial, are not reflected by similarly pronounced differences in response strength among generalist VSNs. There are several, not necessarily mutually exclusive explanations for this finding: First, concentration could simply not be a read-out parameter for VSNs, which would support previous ideas of concentration-invariant VSN activity (Leinders-Zufall et al., 2000). Second, the concentrations in freshly released urine could just exceed the dynamic tuning range of VSNs since, particularly for VOCs, natural signals (e.g., in scent marks) must be accessible to a recipient for a prolonged amount of time (sometimes days). A similar rationale could explain the increased protein concentrations in male urine, since male mice use scent marking to establish and maintain their territories and urinary lipocalins serve as long-lasting reservoirs of VOCs (Hurst et al., 1998). Third, generalist VSNs

might sample information only from a select subset of urinary compounds, which, given their role as biologically relevant chemosignals, might be released at tightly controlled (and thus similar) concentrations. Forth, to some extent, different response profiles could be attributed to non-volatile small organic molecules such as steroids (Nodari et al., 2008), which were beyond the focus of our chemical analysis.”).

(2) How many different wild mouse urines were tested in this study? Is this sufficient to capture the diversity of wild M. musculus in local (Prague) habitats?

We thank the reviewer for pointing this out. For the present study, 20 male (M) and 27 female (F) wild mice were caught at six different sites in the broader Prague area (i.e., Bohnice (50.13415N, 14.41421E; 2M+4F), Dolni Brezany (49.96321N, 14.4585E; 3M+4F), Hodkovice (49.97227N, 14.48039E; 5M+6F), Písnice (49.98988N, 14.46625E; 3M+6F), Lhota (49.95369N, 14.43087E; 1M+2F), and Zalepy (49.9532N, 14.40829E; 6M+5F). 18 of the 27 wild females were caught pregnant. The remaining 9 females were mated with males caught at the same site and produced offspring within a month. When selecting 10 male and 10 female individuals from first-generation offspring for urine collection, we ensured that all six capture sites were represented and that age-matched animals displayed similar weight (~17g). We believe that this capture / breeding strategy sufficiently represents “the diversity of wild M. musculus in local (Prague) habitats.” In the revised manuscript, we have now included these details in the Materials and Methods.

(3) I found Figure 1e and figures in a similar format confusing - one panel describes the response statistics of VSNs, and other panels show the number of compounds found in different MS profiling, which is not immediately obvious from the figures. Is the y-axis legend correct (%)?

We now try make the distinction between VSN “response statistics” and chemical profiling more obvious by gray shadows that link the plots depicting VSN response characteristics to the general pie charts. Moreover, we thank the Reviewer for pointing out the mislabeling of the y-axis. Accordingly, we have deleted “%” in all corresponding figures.

(4) For Figure 5, in order to conclude that the same urine activates a different population of VSNs in two different strains, a critical control is needed to demonstrate that this is not due to the sampling variability - as compositions of V1Rs and V2Rs could vary between different slices, one preferred control is to use VNO slices from the same strain and compare the selectivity used here across the A-P axis.

We thank Reviewer 1 for pointing this out. Importantly, we believe that this is already controlled for (see our response to the Public Review). In fact, for each experiment, we routinely prepare VNO slices along the entire anterior-to-posterior axis (not including the most anterior tip, where the VNO lumen tapers into the vomeronasal duct, and the most posterior part, the lumen “twists” toward the ventral aspect and its volume decreases (see Figs. 7 & S7 in Hamacher et al., 2024, Current Biology)). This usually yields ~7 slices per individual experiment / session. Therefore, we routinely sample and average across the entire VNO anterior-to-posterior axis for each experiment. In Fig. 5, individual independent experiments from each strain (C57BL/6 versus BALB/c) amounted to (a) n = 6 versus n = 8; (b) n = 10 versus n = 10; (c) n = 7 versus n = 9; (d) n = 9 versus n = 10; (e) n = 10 versus n = 9; and (f) n = 12 versus n = 10. Together, we can thus exclude that the considerably different response profiles that we measured using different recipient strains result from a “sampling error.”

To clarify this point in the revised manuscript, we now explain our sampling routine in more detail in the Materials and Methods. Moreover, we now also mention this point in the Results.

Reviewer #2 (Recommendations For The Authors):

(1) Pg 5 Lines 3-16: This summary paragraph contains too much detail given that the reader has not read the paper yet, which makes it bewildering. This should be condensed.

We agree and have substantially condensed this paragraph.

(2) Pg 6 Line 5-8: This summary of the experimental design is obtuse and should be edited for clarity.

We have edited the relevant passage for clarity.

(3) Pg 6 Line 11: "VSNs were categorized..." Specialist vs generalist is defined as responding to one or both stimuli. This definition is placed right after saying that the cells were also tested with KCl. The reader might think that specialist vs generalist was defined in relation to KCl.

We have edited this sentence, which now reads: "Dependent on their individual urine response profiles, VSNs were categorized as either specialists (selective response to one stimulus) or generalists (responsive to both stimuli)."

(4) Pg 6 Line 13: "we recorded urine-dependent Ca²⁺ signals from a total of 16,715 VSNs". Is a "signal" a response? Did all 16,715 VSNs respond to urine? What was the total of KCl responsive cells recorded?

We edited the corresponding passage for clarification. The text now reads: "Overall, we recorded >43,000 K⁺-sensitive neurons, of which a total of 16,715 VSNs (38.4%) responded to urine stimulation. Of these urine-sensitive neurons, 61.4% displayed generalist profiles, whereas 38.6% were categorized as specialists (Figure 1c,d)."

(5) Pg 7 Line 6: The repeated use of the word "pooled" is confusing as it suggests a variation in the experiment. The authors should establish once in the Methods and maybe in the Results that stimuli were pooled across animals. Then they should just refer to the stimulus as male or female or BALB/c rather than "pooled" male etc.

We acknowledge the reviewer's argument. Accordingly, we now introduce the experimental use of pooled urine once in the Methods and in the introductory paragraph of the Results. All other references to "pooled" urine in the Results and Captions have been deleted.

(6) Pg 7 Line 10: "...detected in ≥ 3 out of 10 male..." For the chemical analysis, were these samples not pooled?

Correct. We deliberately did not pool samples for chemical analysis, but instead analyzed all individual samples separately (i.e., 60 samples were subjected to both proteomic and metabolomic analyses). Thus, the criterion that a VOC or protein must be detected in at least 3 of the 10 individual samples from a given sex/strain combination for a 'present' call (and in at least 6 of the 10 samples to be called 'enriched') ensures that the molecular signatures we identify are not "contaminated" by unusual aberrations within single samples. For clarification, we now explicitly outline this procedure in the Methods (Experimental Design and Statistical Analysis – Proteomics and metabolomics).

(7) Pg 7 Line 23: *In line 7, the specialist rate was defined as 5% in reference to the total KCl responsive cells. Here the specialist rate is defined from responsive cells. This is confusing.*

We apologize for the confusion. In both cases, the numbers (%) refer to all K⁺-sensitive neurons. We have added this information to both relevant sentences (l. 7 as well as ll. 23-24). Note that the rate in ll. 23-24 refers to generalists.

(8) Pg 7 Line 25: *Concentration index should be defined before its use here.*

We have revised the corresponding sentence, which now reads: “By contrast, analogously calculated concentration indices (see Materials and Methods) that can reflect potential disparities are distributed more broadly and non-normally (Figure 1h).”

(9) Pg 7 Line 29: *change "trivially" to "simply".*

Done

(10) Pg 7 Line 30: *What is meant by a "generalist" ligand? The neurons are generalists. Probably should read "common ligands"*

We have changed the text accordingly.

(11) Pg 7 Line 31: *What is meant by "global observed concentration disparities" ?*

We have changed the text to “...represented by the observed general concentration disparities.”

(12) Pg 8 Lines 7-11: *This section needs to be edited for clarity as it is very difficult to follow. For example, the definition of "enriched" is buried in a parenthetical. Also, it is very difficult to figure out what a "sample" is in this paper. Is it a pooled stimulus, or is it urine from an individual animal?*

We apologize for the confusion. Throughout the paper a “sample” is a pooled stimulus (from all 10 individuals of a given sex/strain combination) for all physiological experiments. For chemical analysis a “sample” refers to urine from an individual animal.

(13)Pg 8 Line 11: *"abundant proteins" Does this mean absolute concentration or enriched in one sample vs another?*

We changed the term “abundant” to “enriched” as this descriptor has been defined (present in ≥6 of 10 individual samples) in the previous sentence.

(14) Pg 8 Line 18: *"While 32.9% of all..." Please edit for clarity. What is the point?*

The main point here is that, for VOCs, the vast majority of compounds (91.3%) are either generic mouse urinary molecules or are sex/strain-specific.

(15) Pg 10 Line 18: *"Increased VSN selectivity..." This title is misleading as it suggests a change in sensitivity with animal exposure. I think the authors are trying to say "VSNs are more selective for strain than for sex". The authors should avoid the term "exposure to" when they mean "stimulation with" as the former suggests chronic exposure prior to testing.*

We thank the reviewer for the advice and have changed the title accordingly. We also edited the text to avoid the term "exposure to" throughout the manuscript.

(16) Pg 12 Line 10: "we recorded hardly any..." *Hardly any in comparison to what?*
BALB/c?

We apologize for the confusion. We have edited the text for clarity, which now reads: "In fact, (i) compared to an average specialist rate of $11.2\% \pm 6.6\%$ (mean $\pm$ SD) calculated over all 13 binary stimulus pairs ($n = 26$ specialist types), we observed only few specialist responses upon stimulation with urine from wild females (2% and 3%, respectively), and..."

Reviewer #3 (Recommendations For The Authors):

(1) *Related to the pairwise stimulus-response experimental design and analysis: there is precedent in the field for studies that explore the same topic (sex- and strain-selectivity), but measure VSN sensitivity across many urine stimuli, not just two at a time. This has been done both in the VNO (He et al, Science, 2008; Fu, et al, Cell, 2015) and in the AOB (Tolokh, et al, Journal of Neuroscience, 2013). The current manuscript does not cite these studies.*

Reviewer 3 is correct and we apologize for this oversight. We now cite the two VSN-related studies by He et al. and Fu et al. in the Introduction.

(2) The findings of the mass spectrometry-based profiling of mouse urine - especially for volatiles - is only accessible through repositories, making it difficult to for readers to understand which molecules were found to be highly divergent between sexes/strains. There is value in the list of ligands to further investigate, but this information should be made more accessible to readers without having to comb through the repositories.

We agree that there "is value in the list of ligands to further investigate" and, accordingly, we now provide a table (Table 1) that lists the top-5 VOCs that – according to sPLS-DA – display the most discriminative power to classify samples by sex (related to Figure 2c) or strain (related to Figure 2d). For ease of identification, all entries list internal mass spectrometry identifiers, identifiers extracted from MS analysis database, the sex or strain that drives separation, which two-dimensional component / x-variate represents the most discriminative variable, PubChem chemical formula, PubChem common or alternative names, Chemical Entities of Biological Interest or PubChem Compound Identification, and the VOC's putative origin.

(3) *There is a long precedent for integrating molecular assessments and physiological recordings to identify specific ligands for the vomeronasal system:*

- *nonvolatiles (e.g., Leinders-Zufall, et al., Nature, 2000)*
- *peptides (e.g., Kimoto et al., Nature, 2005; Leinders-Zufall et al. Science, 2004; Riviere et al., Nature, 2009; Liberles, et al., PNAS, 2009)*
- *proteins (e.g., Chamero et al., Nature, 2007; Roberts et al., BMC Biology, 2010)*
- *excreted steroids and bile acids (Nodari et al., Journal of Neuroscience, 2008; Fu et al., Cell, 2015; Doyle, et al., Nature Communications, 2016)*

The Leinders-Zufall (2000), Roberts, and Nodari papers are referenced, but the broader efforts by the community to find specific drivers of vomeronasal activity are not fully represented in the manuscript. The focus of this paper is fully related to this broader

effort, and it would be appropriate for this work to be placed in this context in the introduction and discussion.

We now refer to all of the studies mentioned in the Introduction (except the article published by Liberles et al. in 2009, since the authors of that study do not identify vomeronasal ligands).

(4) Throughout the manuscript (starting in Fig. 1h) the figure panels and captions use the term "response index" whereas the methods define a "preference index." It seems to be the case that these two terms are synonymous. If so, a single term should be consistently used. If not, this needs to be clarified.

We now consistently use the term “response index” throughout the manuscript.

(5) It would be useful to provide a table associated with Figure 2 - figure supplement 1 that lists the common names and/or chemical formulas for the volatiles that were found to be of high importance.

We agree and, accordingly, we now provide a table (Table 2) that lists VOC, which – according to Random Forest classification and resulting Gini importance scores – display the most discriminative power to classify samples by sex (related to Figure 2 - figure supplement 1a) or strain (related to Figure 2 - figure supplement 1b). Notably, it is generally reassuring that several VOCs are listed in both Table 1 and Table 2, emphasizing that two different supervised machine learning algorithms (i.e., sPLS-DA (Table 1) and Random Forest (Table 2)) yield largely congruent results.

(5) The use of the term "comprehensive" for the molecular analysis is a little bit misleading, as volatiles and proteins are just two of the many categories of molecules present in mouse urine.

We have now deleted most mentions of the term "comprehensive" when referring to the molecular analysis.

(7) Page 11, lines 24-27: The sentences starting "We conclude..." and ending in "semiochemical concentrations." These two sentences do not make sense. It is not known how many of the identified proteins are actual VSN ligands. Moreover, there is abundant evidence from other studies that individual VSN activity provides information about distinct semiochemical concentrations.

We have substantially edited and rephrased this paragraph to better reflect that different scenarios / interpretations are possible. The relevant text now reads: “We conclude that VSN population response strength might not be so strongly affected by strain-dependent concentration differences among common urinary proteins. In that case, it would appear somewhat unlikely that individual VSN activity provides fine-tuned information about distinct semiochemical concentrations. Alternatively, as some (or even many) of the identified proteins could not serve as vomeronasal ligands at all, generalist VSNs might sample information from only a subset of compounds which, in fact, are secreted at roughly similar concentrations.”

(8) The explanation of stimulus timing is mentioned several times but not defined clearly in methods. Page 19, lines 14-19 have information about the stimulus delivery device, but it would be helpful to have stimulus timing explicitly stated.

In addition to the relevant captions, we now explicitly state stimulus timing (i.e., 10 s stimulations at 180 s inter-stimulus intervals) in the Results.

(9) Typos: Page 10, line 7: "male biased" → "male-biased" for clarity

Wilcoxon "signed-rank" test is often misspelled "Wilcoxon singed ranked test" or "Wilcoxon signed ranked test"

In the Fig. 3 legend, the asterisk meaning is unspecified.

"(im)balances" → imbalances (page 27, line 24; page 37, line 16; page 38, line 16)

Figure 2 - figure supplement 1 and in Figure 2 - figure supplement 2, in the box-andwhisker plots the units are not specified in the graph or legend."

We have made all required corrections.